Group Assignment 2: Feature Selection

Statement of Contribution

Feature selection methods

- Sebastian: Relieff
- Gary: fscchi2
- Lenard: fscmrmr

Deliverable	Sebastian	Gary	Lenard
Generating t-SNE plots	All	All	All
Hierarchical clustering	All	All	All
Feature Selection	relieff	fscchi2	fscmrmr
Predictor Importance Scores	relieff	fscchi2	fscmrmr
Classification Learner App (Machine Learning)	relieff	fscchi2	fscmrmr
Debugging	All	All	All

By submitting this assignment all group members agree to the above listed contributions to be true and accurate.

Report

A t-SNE plot was created to visualize the miRNA of samples included in the Gastrointestinal Neuroendocrine (GI-NET) cancer dataset by type, involving pancreatic, ileal, appendiceal, and rectal.

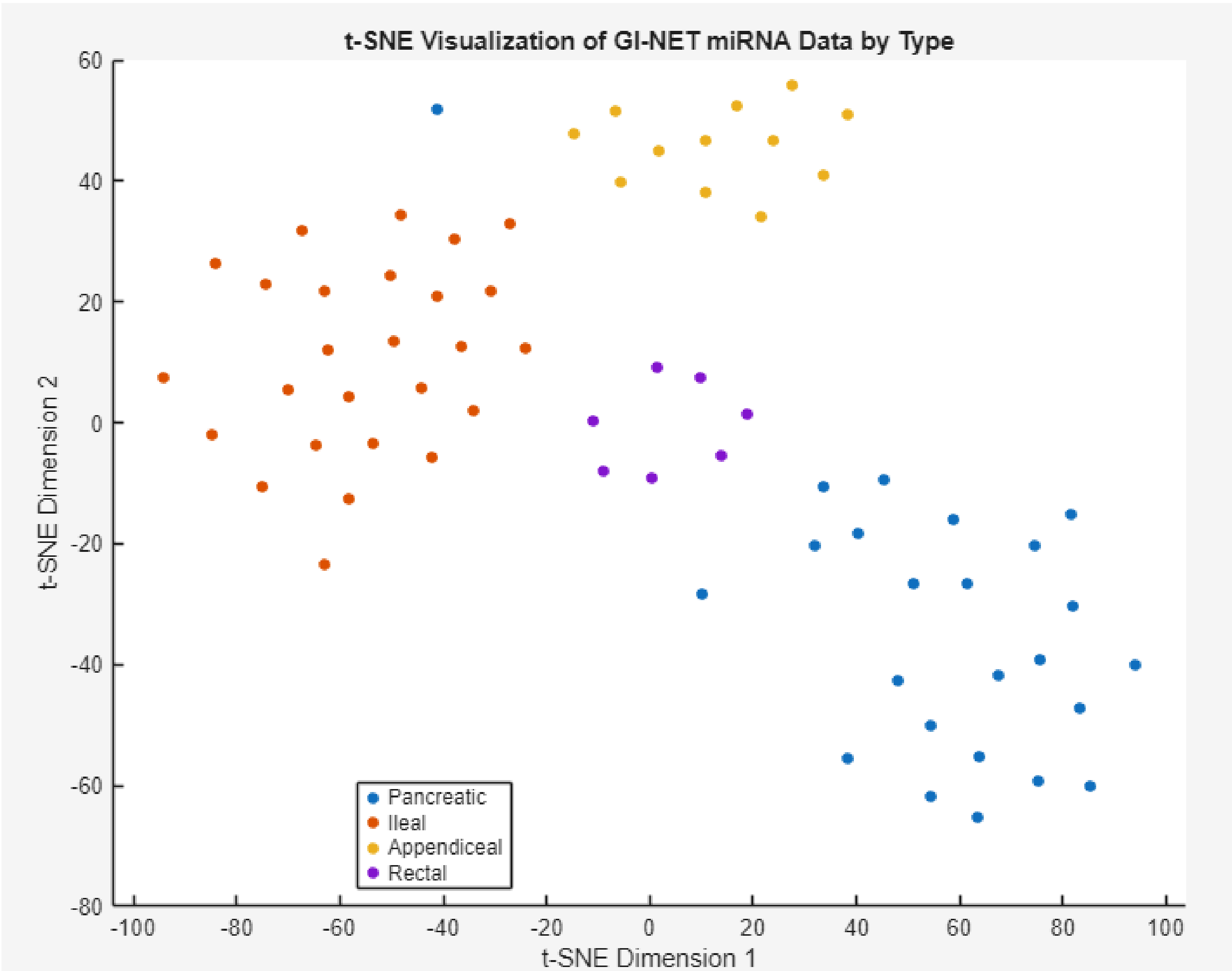

Figure 1. t-SNE visualization with barneshut algorithm, spearman distance, and perplexity set to 20.

Figure 1 illustrates that samples cluster together depending on which of the four cancer types that they have. One sample with pancreatic cancer does not cluster well together with other pancreatic cancer samples, being mapped near the appendiceal and ileal samples.

Hierarchical clustering was performed on all of the miRNA measured in the GI-NET cancer dataset, investigating the clustering of samples based on either cancer type or cancer grade.

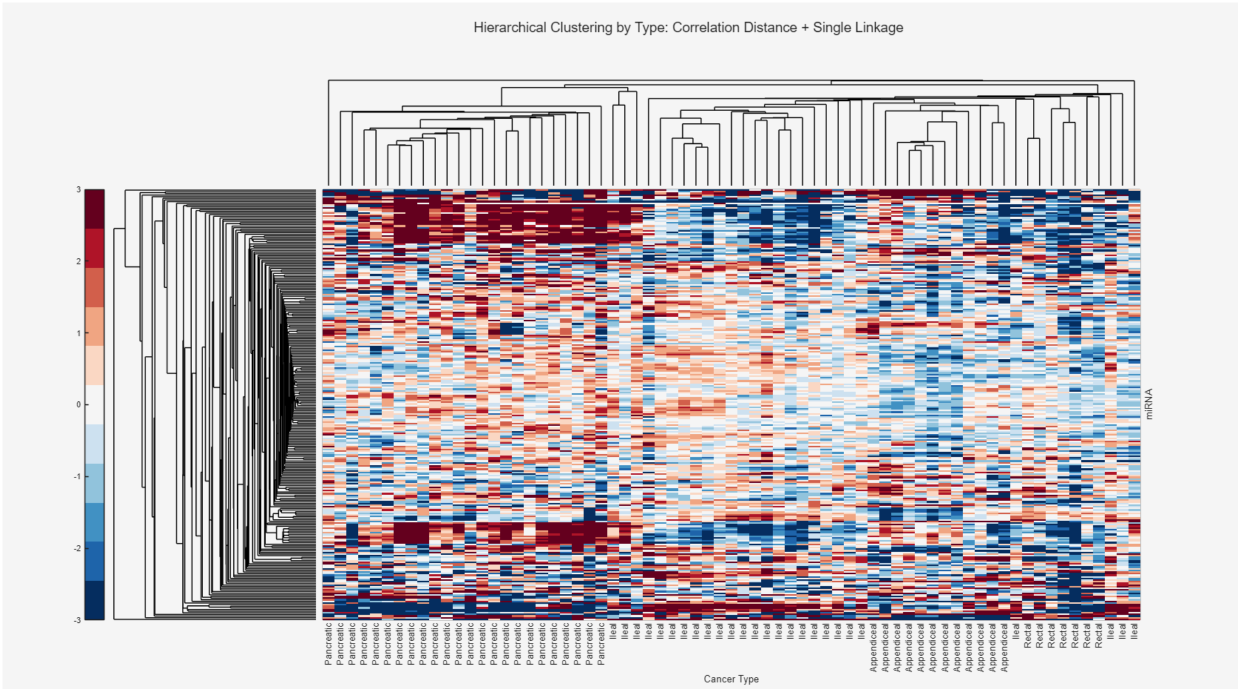

Figure 2. Hierarchical clustering of samples by cancer type (pancreatic, ileal, appendiceal, and rectal) using correlation distance and single linkage.

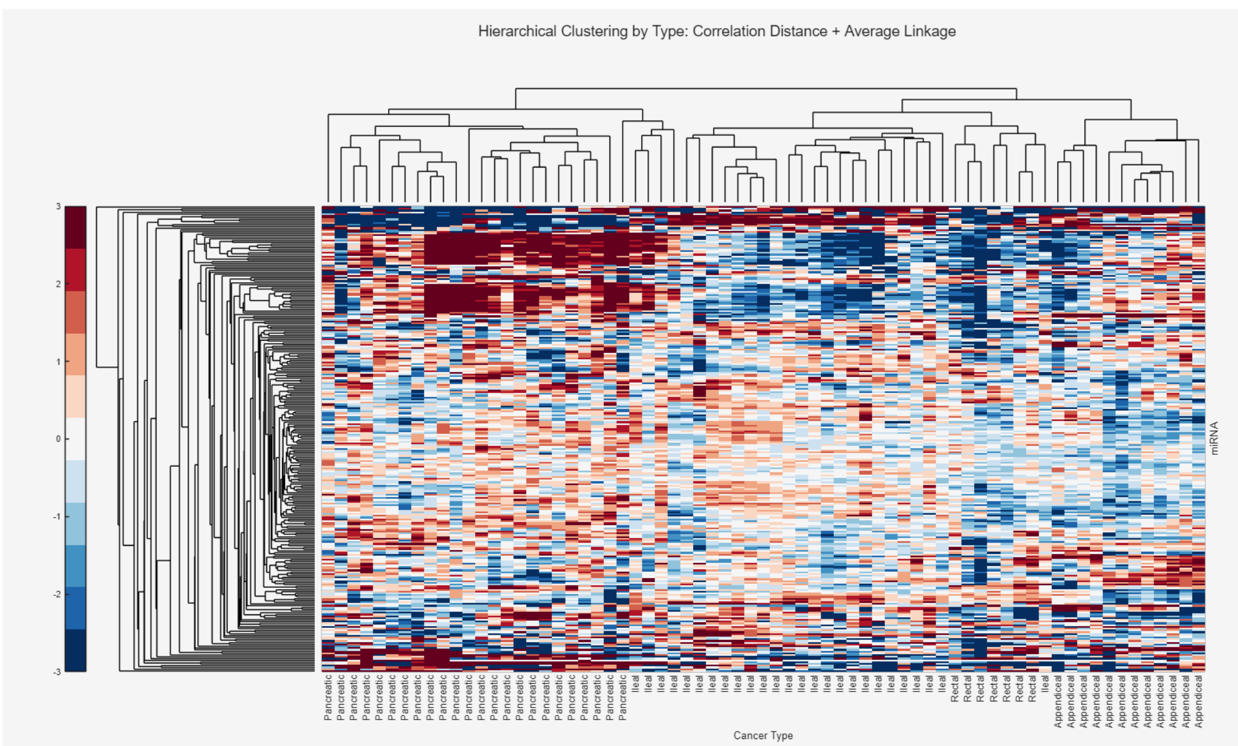

Figure 3. Hierarchical clustering of samples by cancer type (pancreatic, ileal, appendiceal, and rectal) using correlation distance and average linkage.

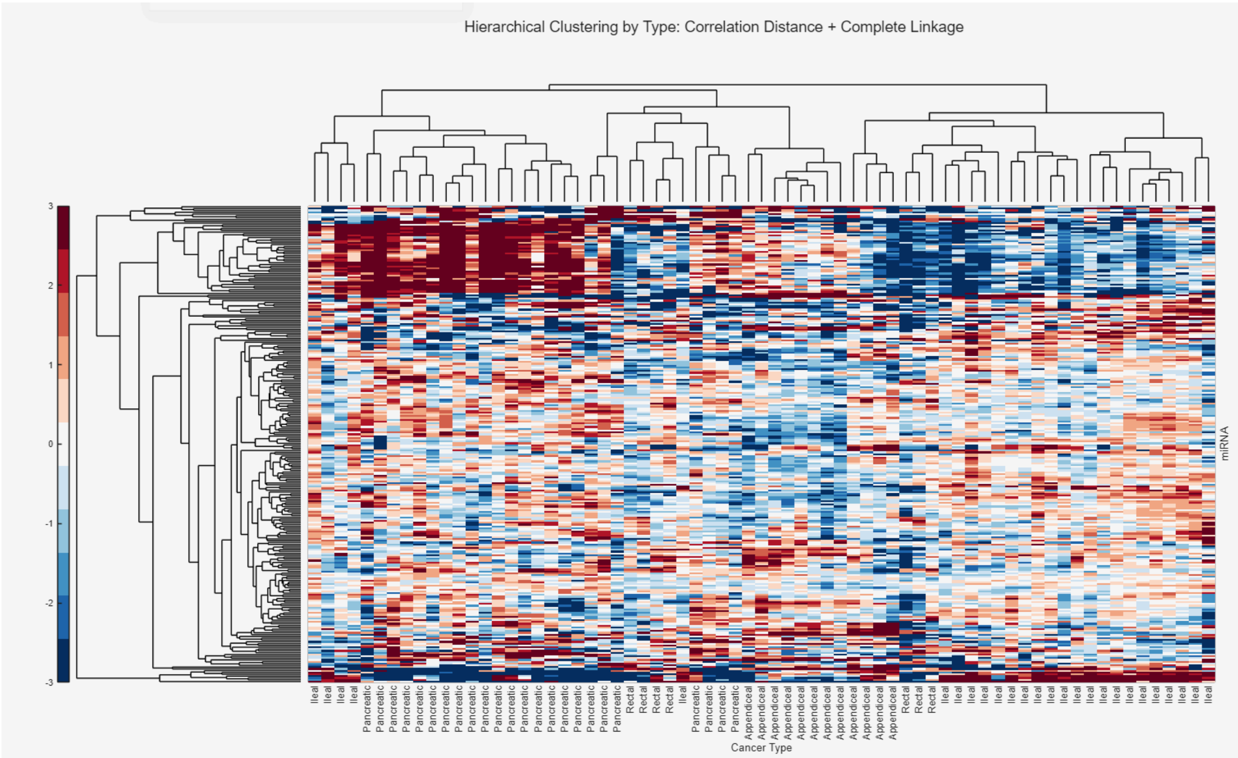

Figure 4. Hierarchical clustering of samples by cancer type (pancreatic, ileal, appendiceal, and rectal) using correlation distance and complete linkage.

Figures 2, 3, and 4 illustrate hierarchical clustering of samples by cancer type. Using correlation distance as a similarity measure, each of the three graphs illustrate different linkages being single, average, or complete linkages respectively. In both single and average linkage methods, the samples cluster well together based on cancer type and only a few samples do not form clusters with other samples of similar type. There is slightly more variation in the clustering of samples based on complete linkage. There is some grouping in the heatmap of highly expressed miRNA in samples with pancreatic cancer. Nonetheless, there is overall very little clustering of miRNA expression, which could indicate features with little discriminatory power.

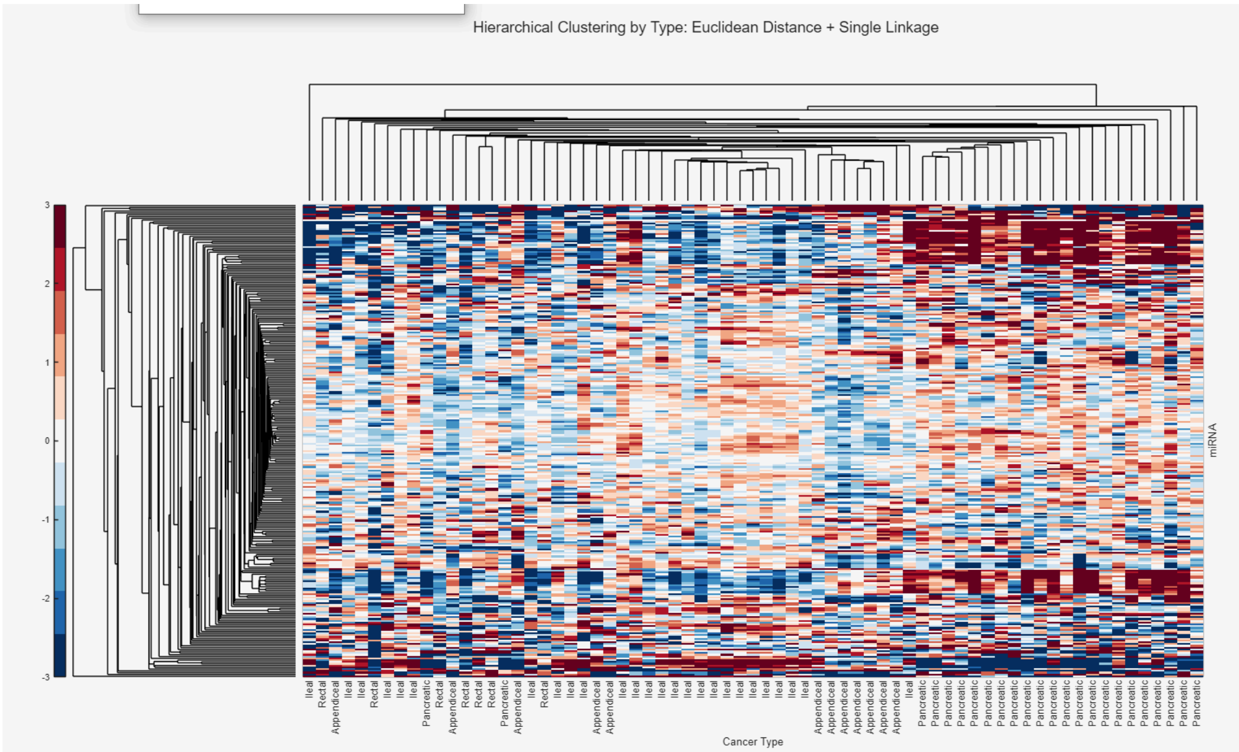

Figure 5. Hierarchical clustering of samples by cancer type (pancreatic, ileal, appendiceal, and rectal) using euclidean distance and single linkage.

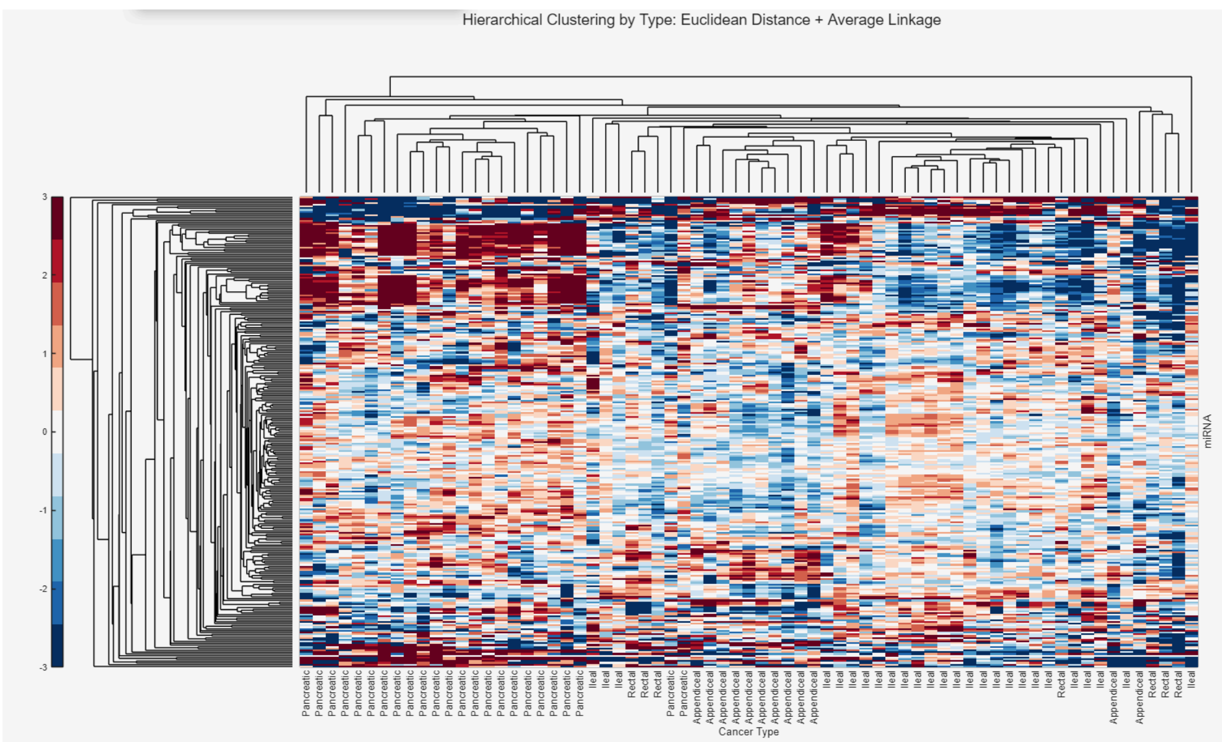

Figure 6. Hierarchical clustering of samples by cancer type (pancreatic, ileal, appendiceal, and rectal) using euclidean distance and average linkage.

Similarly, figures 5 and 6 illustrate hierarchical clustering of samples by cancer type. Nonetheless, euclidean distance was selected as a similarity measure. When investigating clustering between samples based on either single or average linkage measures, both methods resulted in samples being in different clusters than those with a similar sample cancer type. Similar to figures above using correlation distance, there is very little clustering in the heatmap besides a slight grouping of highly expressed miRNA within samples with pancreatic cancer

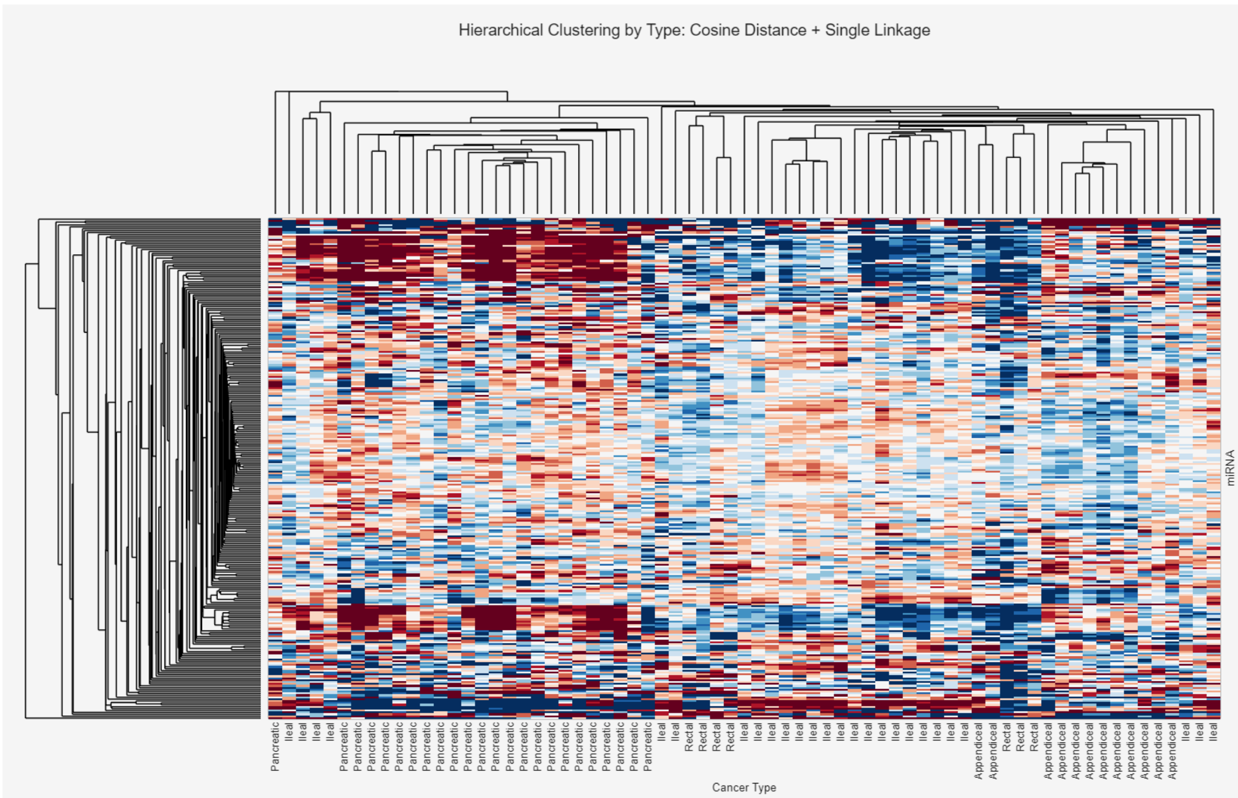

Figure 7. Hierarchical clustering of samples by cancer type (pancreatic, ileal, appendiceal, and rectal) using cosine distance and single linkage.

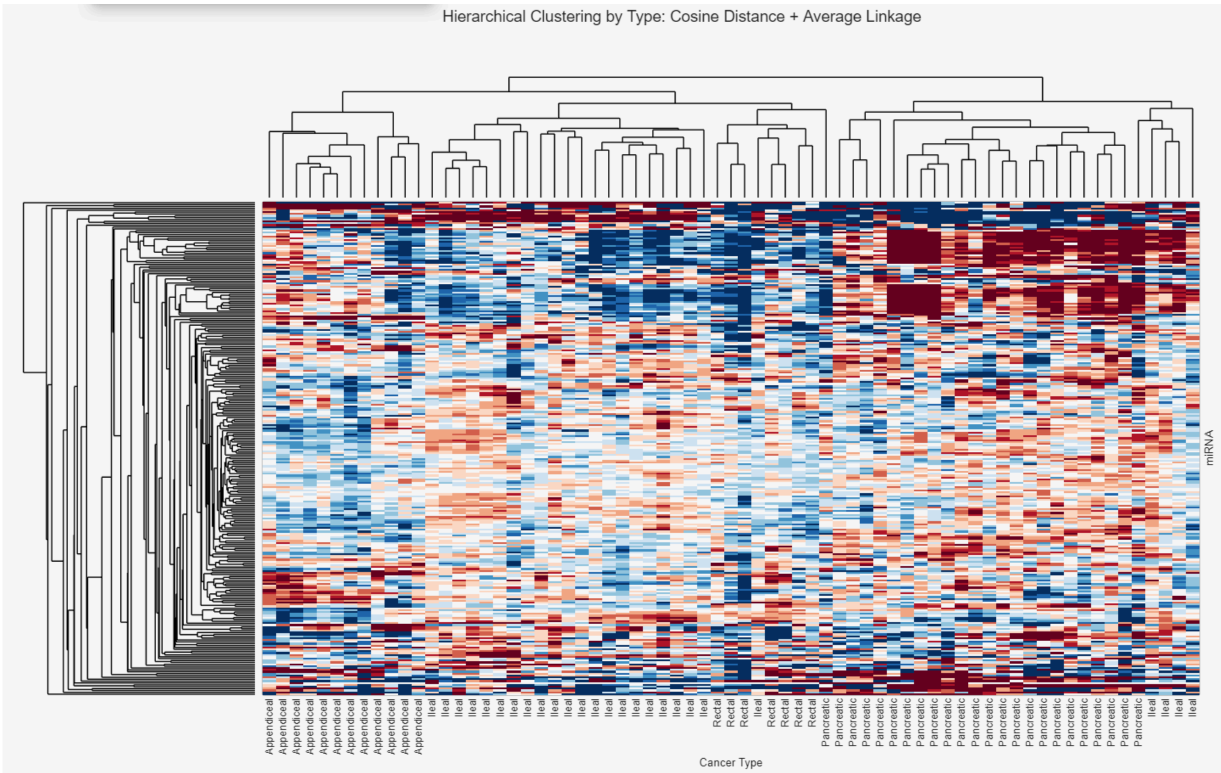

Figure 8. Hierarchical clustering of samples by cancer type (pancreatic, ileal, appendiceal, and rectal) using cosine distance and average linkage.

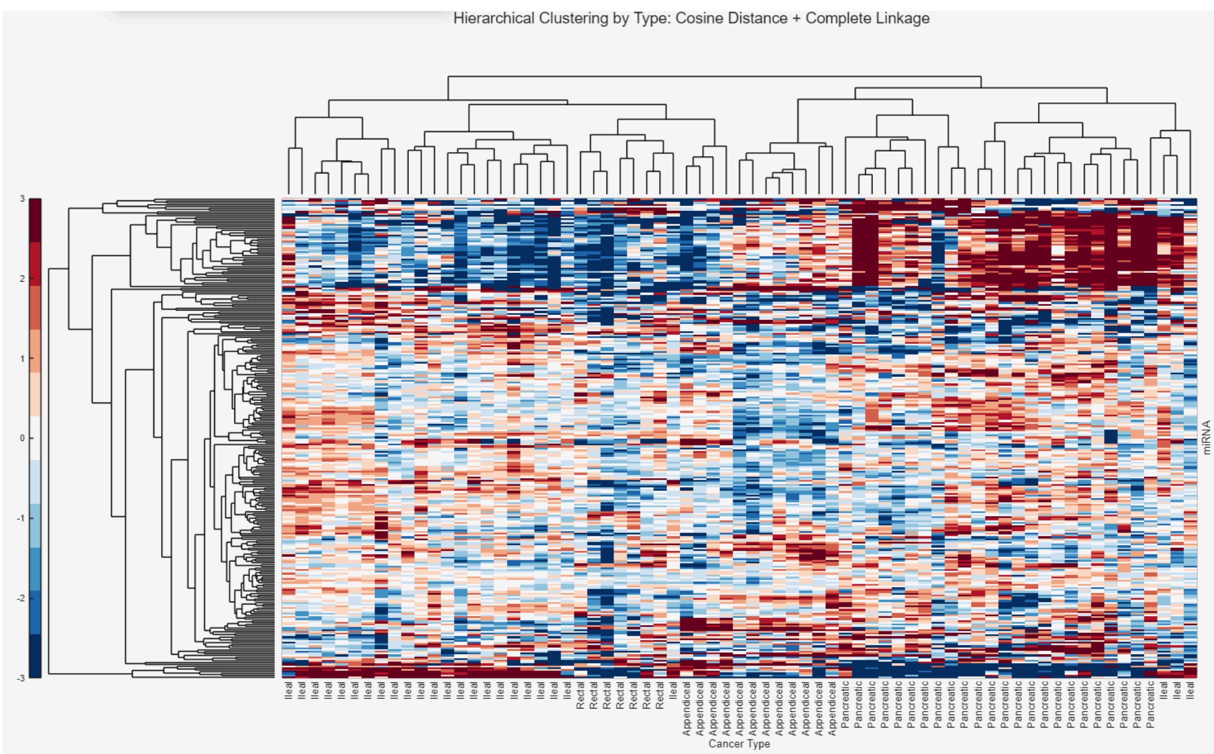

Figure 9. Hierarchical clustering of samples by cancer type (pancreatic, ileal, appendiceal, and rectal) using cosine distance and complete linkage.

Figures 7, 8, and 9 illustrate hierarchical clustering of samples by cancer type using cosine distance as a similarity measure. In each of the three linkage methods, there are some samples that do not cluster with other samples in the dataset of a similar cancer type.

Based on our analysis of the figures above, we decided to stick with the default similarity measure as all clustergrams looked very similar to one another and there were no significant changes. Moreover, average linkages were chosen to prevent the possibility of outliers hindering the clustering of samples.

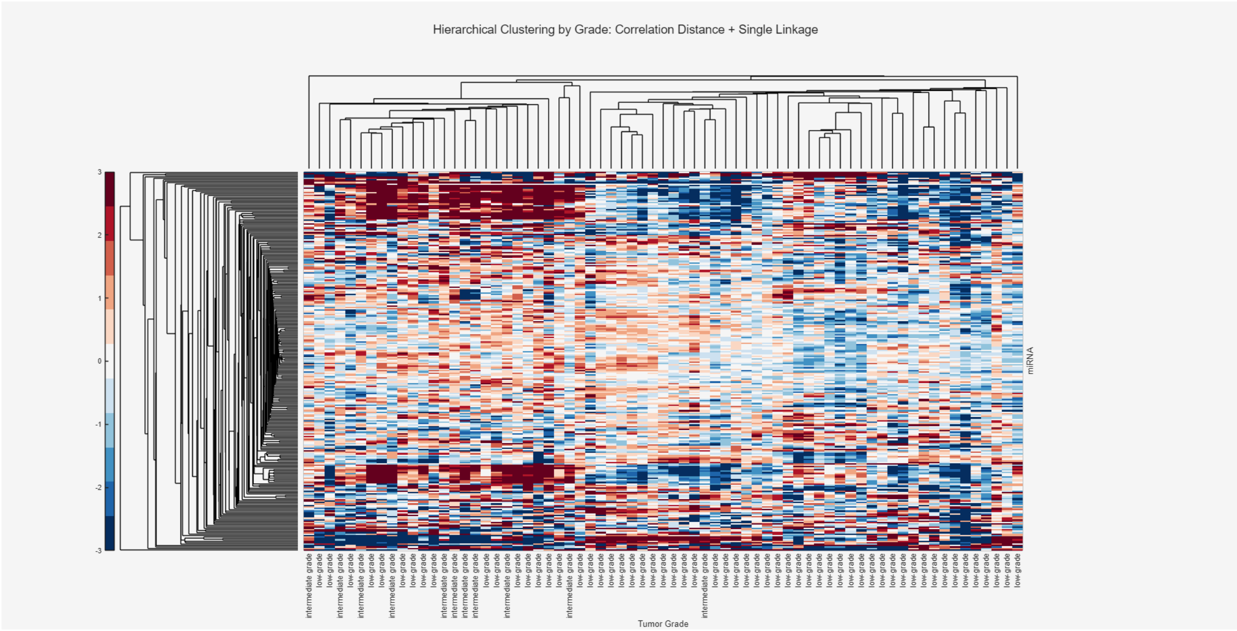

Figure 10.

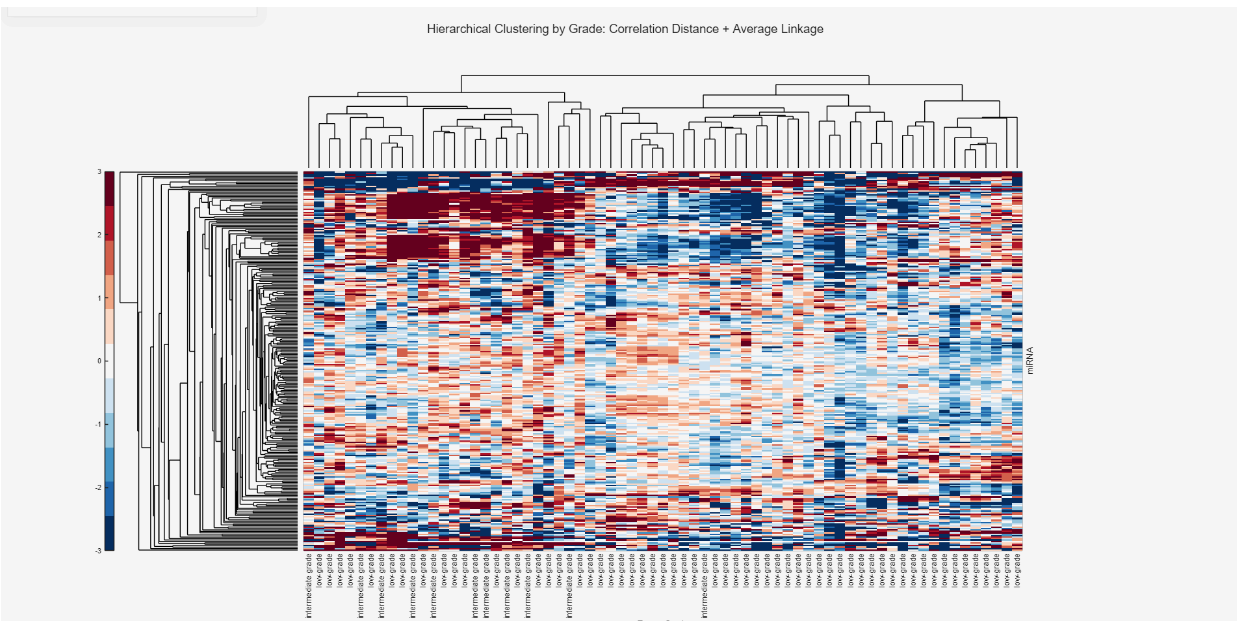

Figure 11.

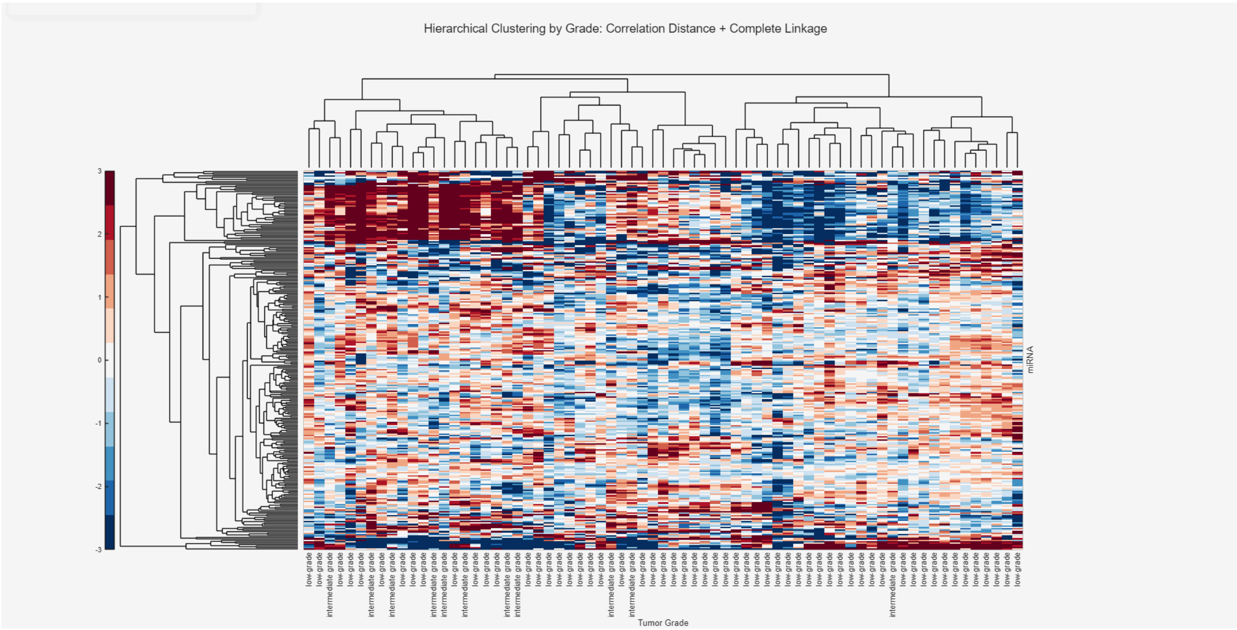

Figure 12.

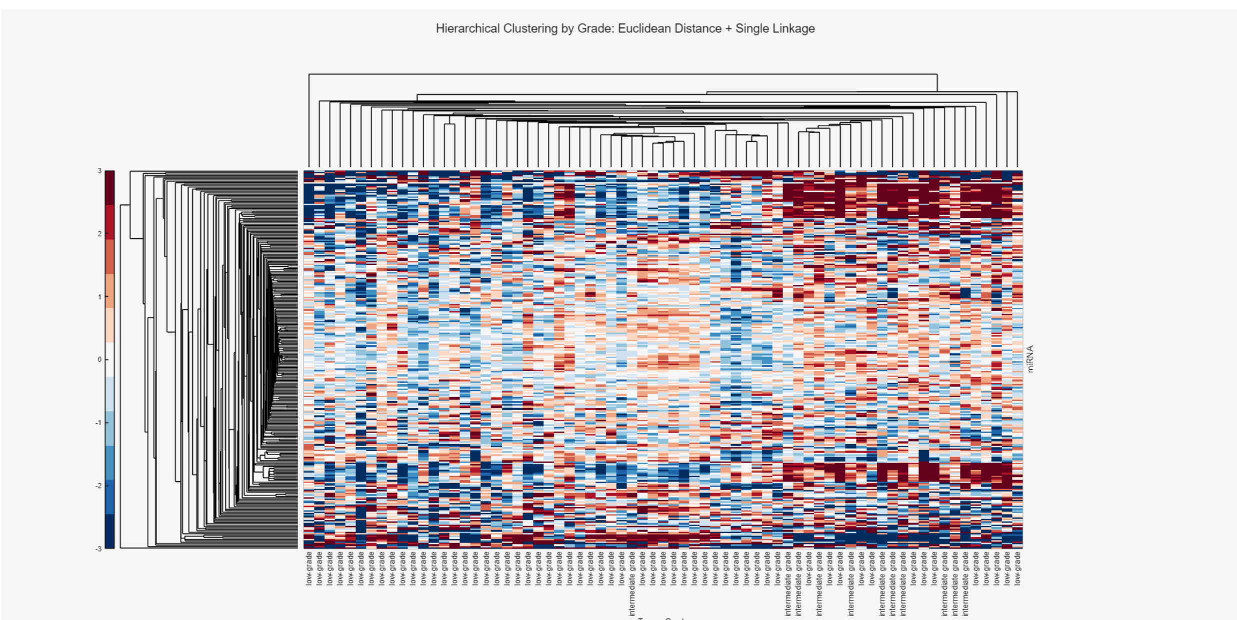

Figure 13.

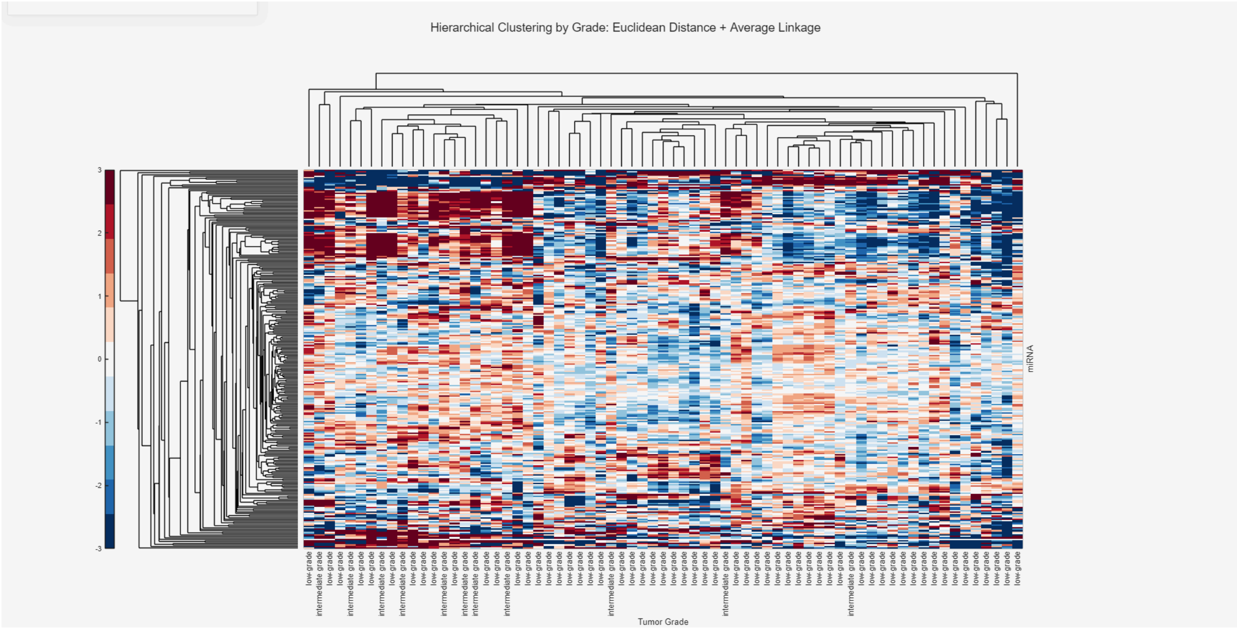

Figure 14.

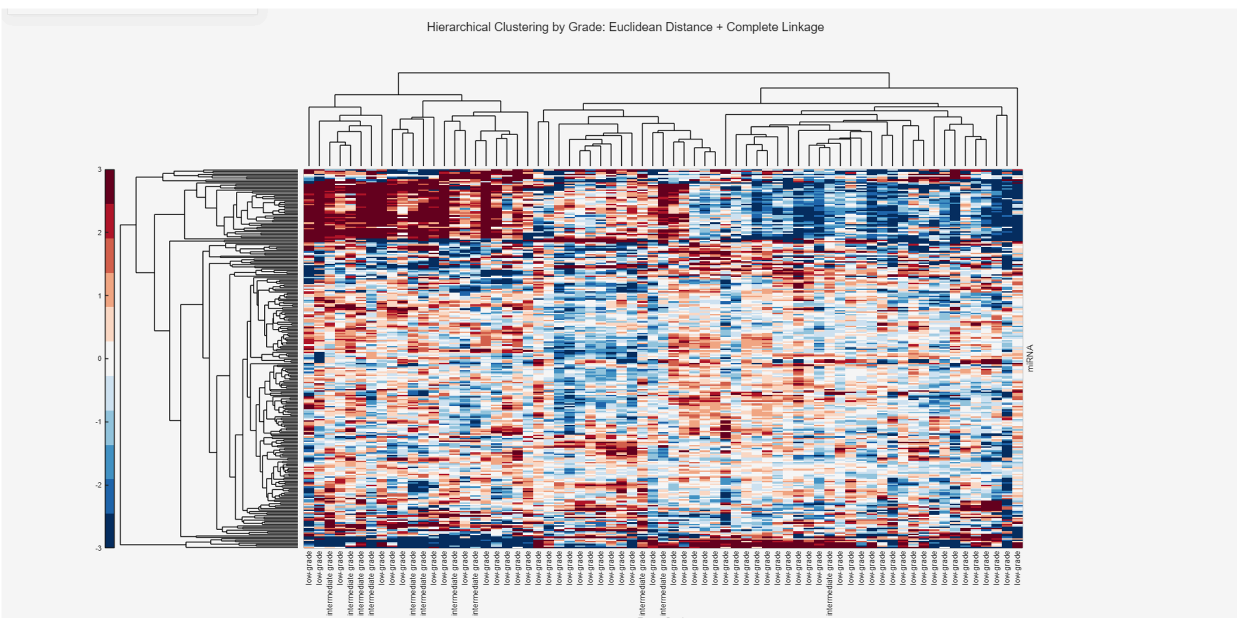

Figure 15.

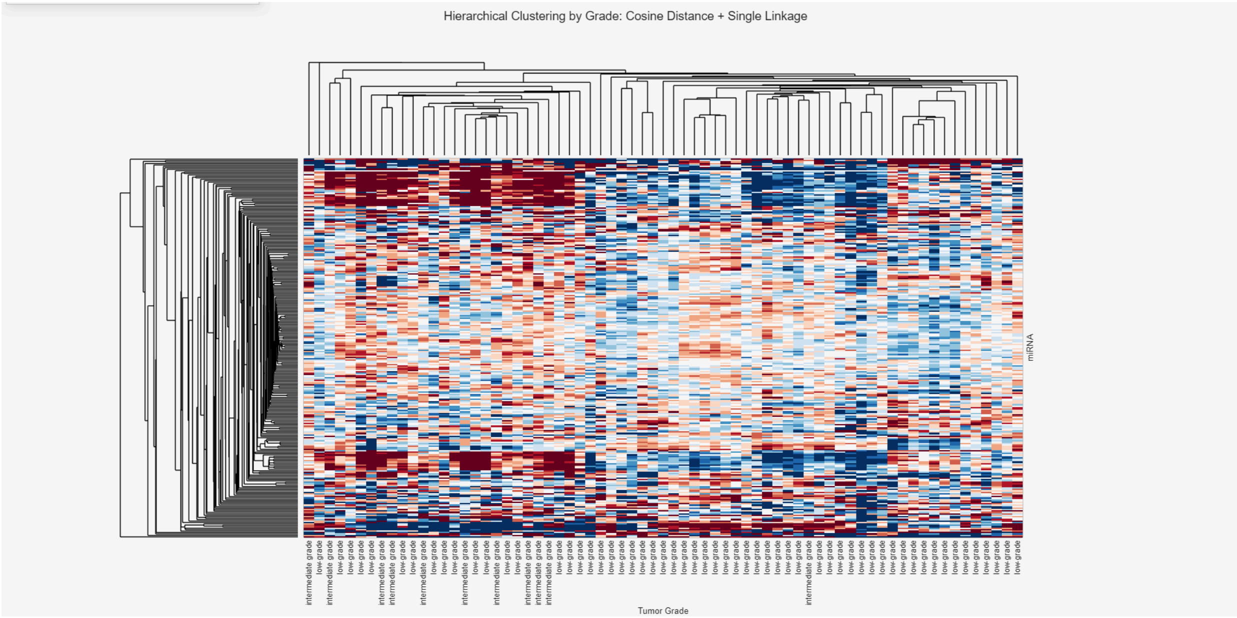

Figure 16.

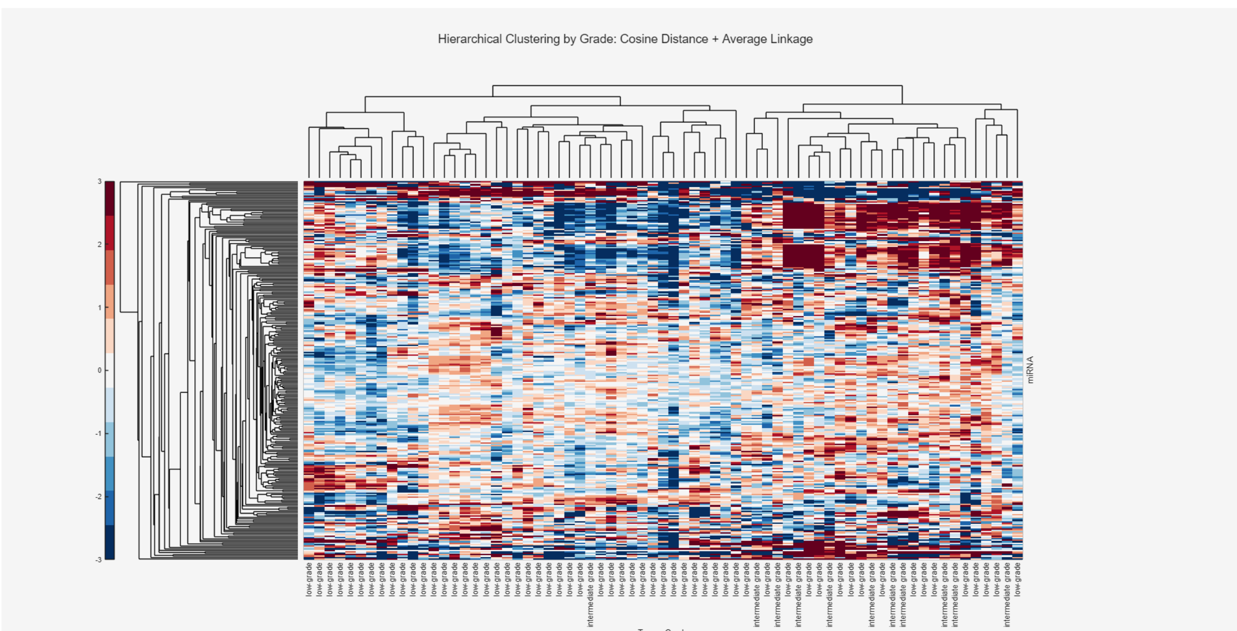

Figure 17.

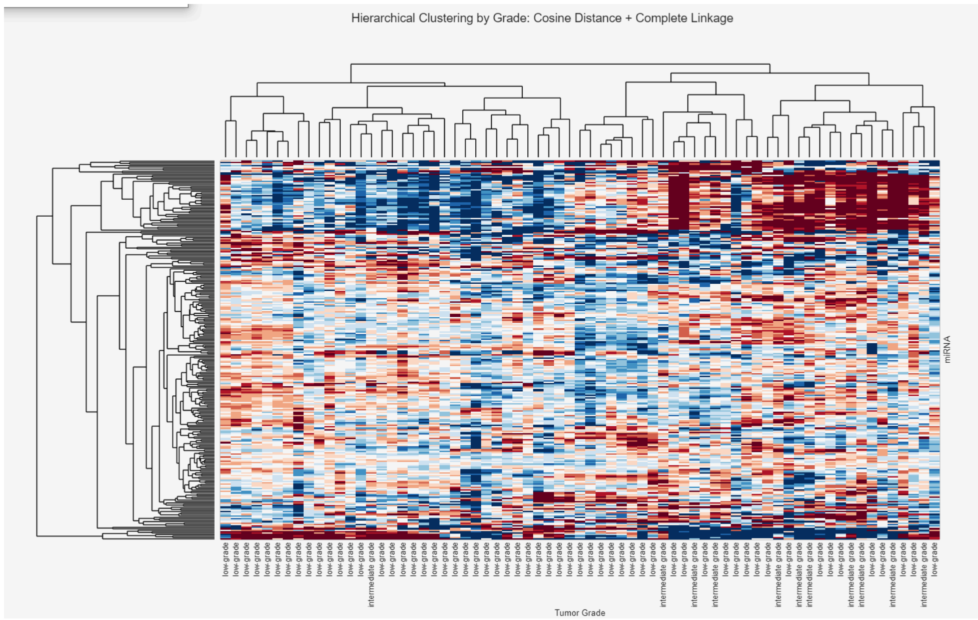
Hierarchical Clustering by Grade: Cosine Distance + Complete Linkage
miRNA
Tumor Grade

Figure 18.

Relieff Feature Selection Method

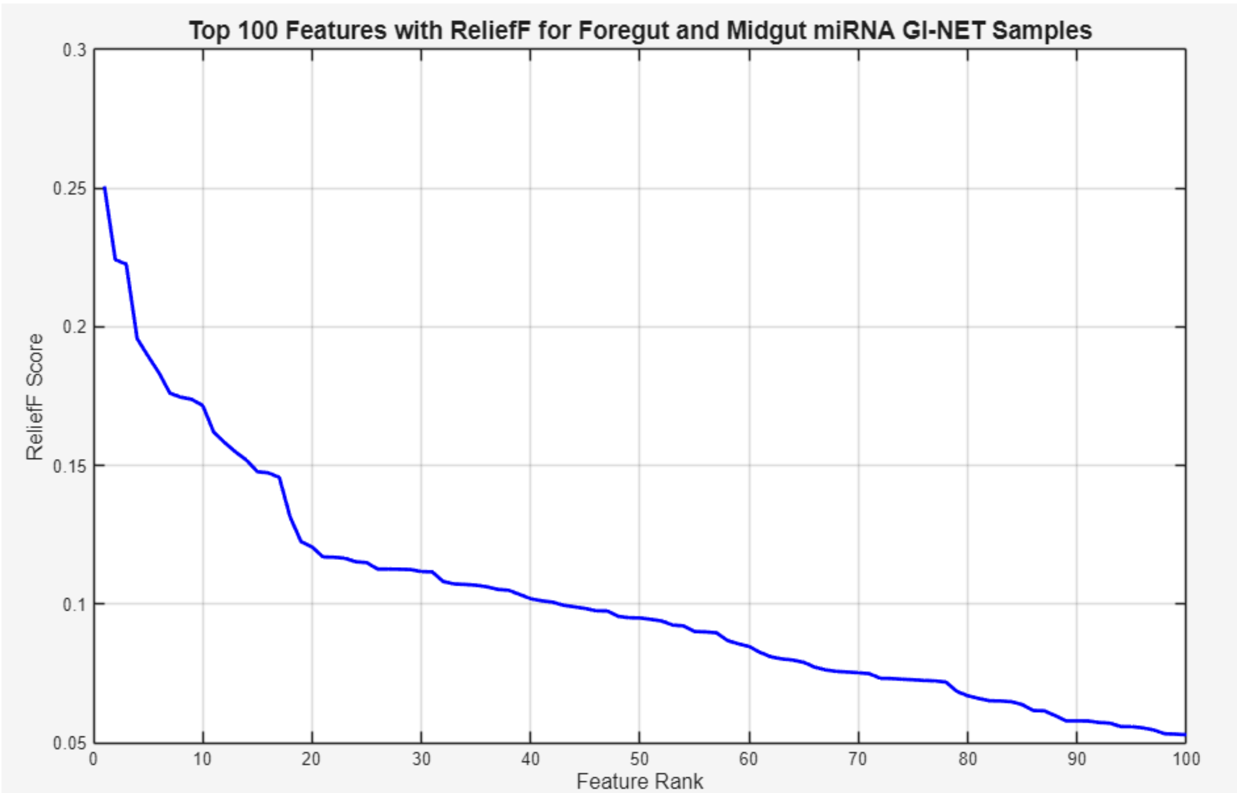

Figure 19. General view of the top 100 features to narrow down selection.

This figure shows a general overview of the top 100 features and their feature importance. From my analysis, it appears there is some elbow and sharp dropoff near 10-12, so I believe it may be effective to use these top 10 features when passing into our machine learning models at a later stage. This may potentially balance the use of enough appropriate features for stronger discrimination while also limiting overfitting.

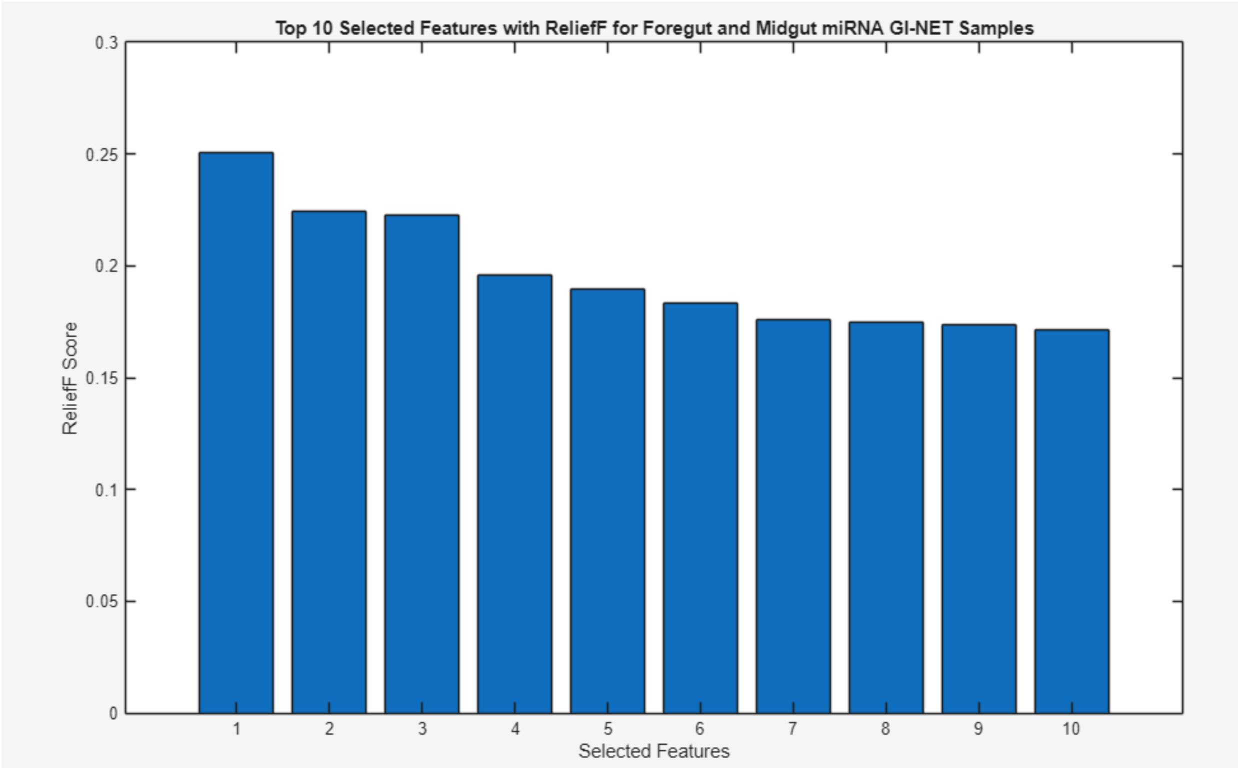

Figure 20. Feature importance of top selected features.

Plotting the top 10 selected features, there is a relatively strong contribution from each of these features. We will use these features when passing them into our models.

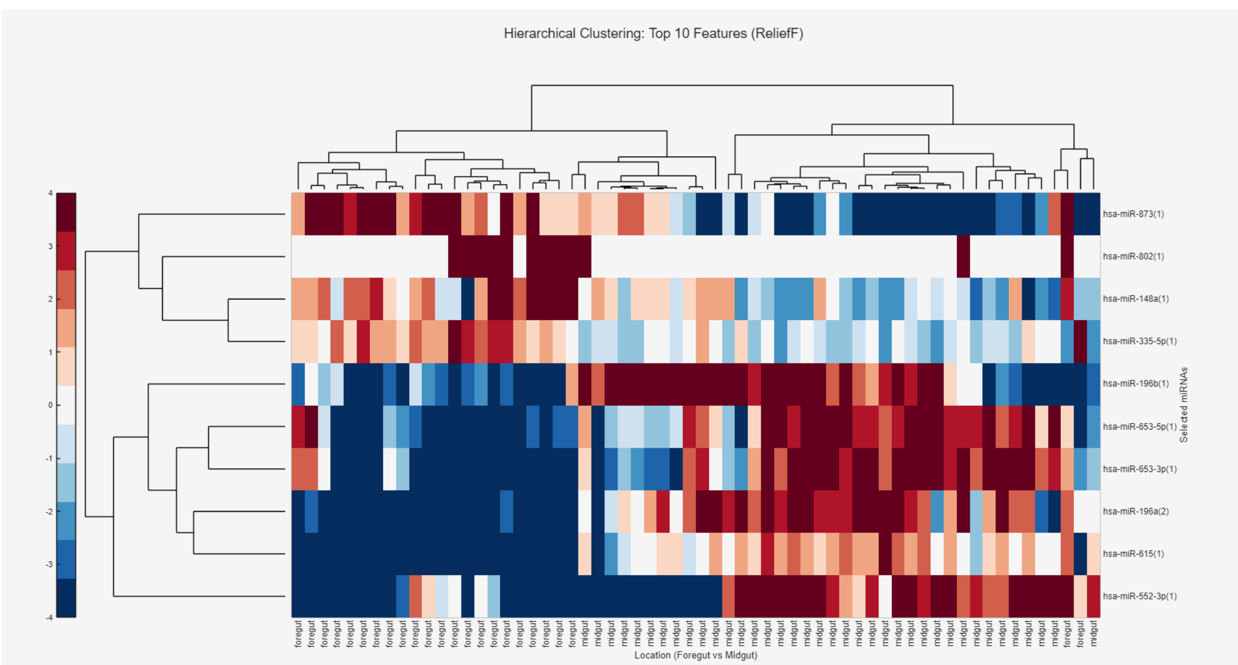

Figure 21. Hierarchical clustering and heatmap of the top 10 selected features, showing improvement in clustering.

In the produced hierarchical clustering and heatmap of the top 10 features, we can clearly see clustering between miRNAs such as miR-196a, miR0615 and miR-522 which show the strongest differential expression between locations. Our early clustering in step 2 used all 300+ miRNAs which introduced lots of noise and did not produce a very good clustering overall, where we had scattered patterns with no clear blocks. Here, we can see distinct bands such as hsa-miR-196b with clear blue foregut samples and clear red midgut samples, which demonstrates better discrimination. This ultimately allowed us to capture essential information, reduce the dimensionality curse, and improve the overall interpretability of our findings.

With 10 features selected, we now move on to our machine learning portion. 5-fold cross-validation was used to give a solid estimate of model performance regarding overfitting, where the dataset is divided into k equal parts and trained on k-1 folds and validated on the remaining one, repeating the process. k= 5 is a solid balance between performance and variance reduction. An 80/20 split was used with 80% on the training data and 20% for the validation hold-out so the models have enough samples to learn patterns while also maintaining decent generalizability.

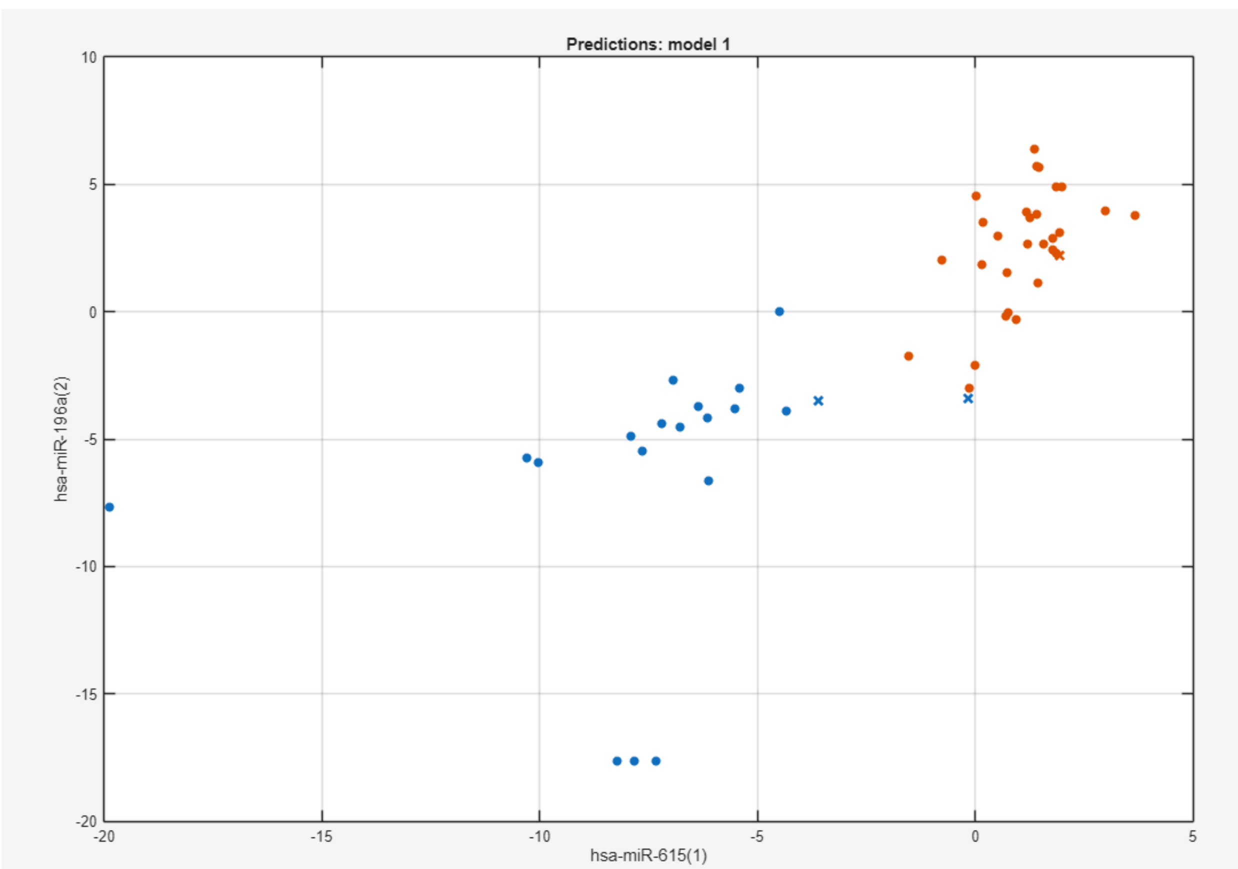

Figure 22. Initial scatterplot setup of Model 1 (Tree with Fine Tree) when loading in the prepared data. Foregut is in blue and midgut is in red.

As we can see from the initial loading above., hsa-miR-615 and hsa-miR-196a appear to have the best group clustering compared to the other predictors.

Table 1. Classification Model Results with relieff feature selection, with Neural Network performing the best with 98% accuracy and Ensemble (PCA) performing the worst with 62% accuracy.

Model Type	Validation Accuracy
Neural Network	98.00%
Discriminant	96.00%
Binary GLM Logistic Regression	96.00%
SVM	96.00%
KNN	96.00%
Ensemble	96.00%
Neural Network	96.00%
Kernel	96.00%
Tree	94.00%
Discriminant	94.00%
Naive Bayes	94.00%
SVM	94.00%
KNN	94.00%
Ensemble	94.00%
Neural Network	94.00%
Kernel	94.00%
SVM (PCA)	94.00%

Neural Network (PCA)	94.00%
Efficient Linear SVM	92.00%
Naive Bayes	92.00%
SVM (PCA)	92.00%
KNN (PCA)	92.00%
Ensemble (PCA)	92.00%
Neural Network (PCA)	92.00%
Efficient Logistic Regression	90.00%
Tree (PCA)	90.00%
Discriminant (PCA)	90.00%
Efficient Logistic Regression (PCA)	90.00%
Efficient Linear SVM (PCA)	90.00%
Naive Bayes (PCA)	90.00%
SVM (PCA)	90.00%
Ensemble (PCA)	90.00%
Neural Network (PCA)	90.00%
Discriminant (PCA)	88.00%
Binary GLM Logistic Regression (PCA)	88.00%
Neural Network (PCA)	88.00%
Kernel (PCA)	88.00%
Naive Bayes (PCA)	86.00%
KNN (PCA)	86.00%

Kernel (PCA)	86.00%
Ensemble	84.00%
KNN (PCA)	84.00%
Ensemble (PCA)	82.00%
SVM (PCA)	72.00%
SVM	64.00%
KNN	62.00%
Ensemble	62.00%
SVM (PCA)	62.00%
KNN (PCA)	62.00%
Ensemble (PCA)	62.00%

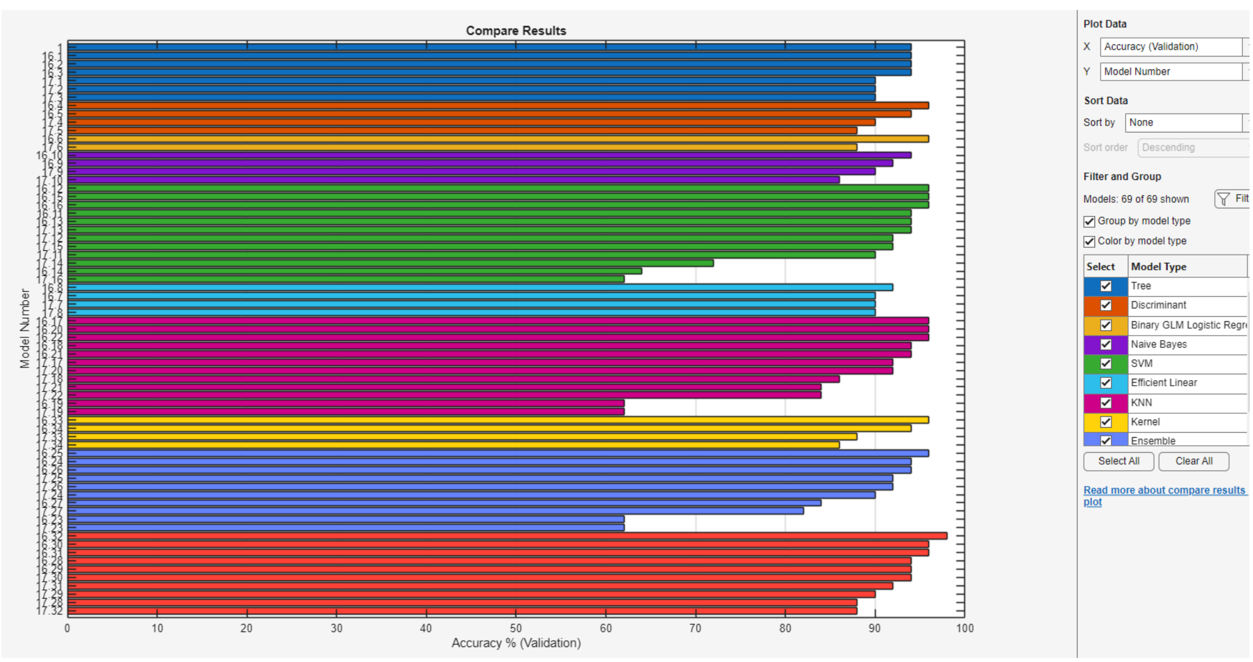

Figure 23. Validation accuracy comparison across selected model types and their groups, with most in the 94%+ range. Ensemble (light purple) and Naive Bayes (dark purple) groups tended to perform the worst on average while Neural Networks (red) and Trees (blue) performed quite well, amongst others.

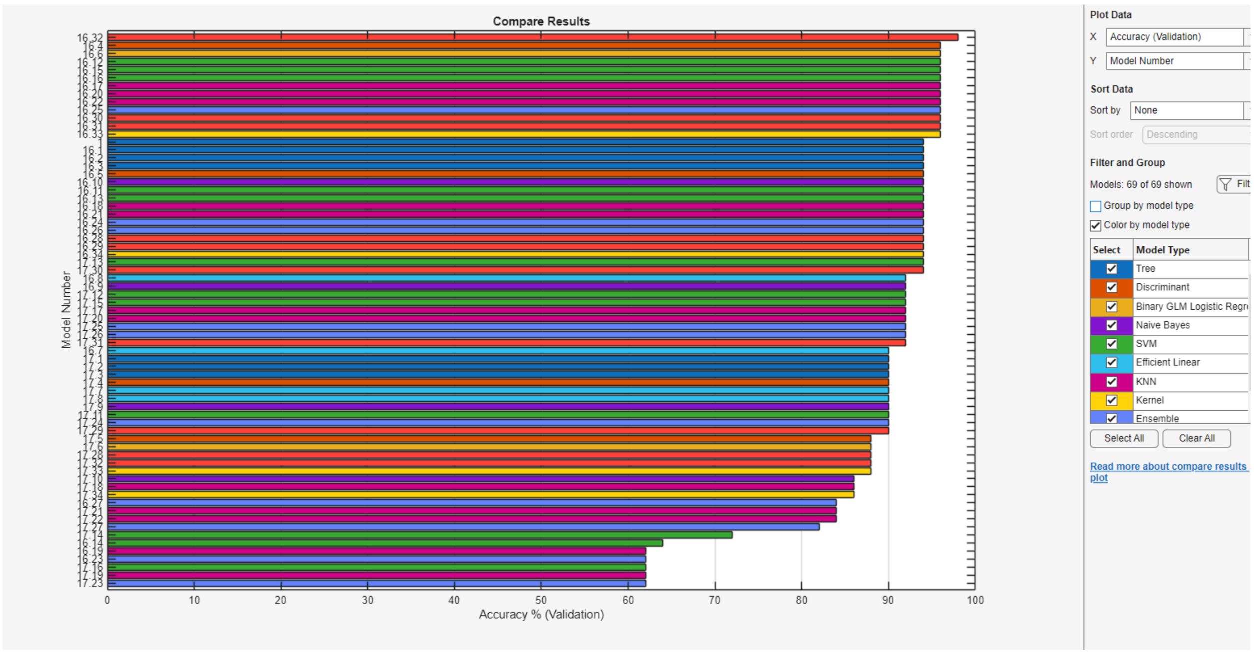

Figure 24. Validation Accuracy Comparison Across all Model Types and Configurations (No PCA, PCA, all 10 features, 6 features). The highest performing model was the Neural Network (trilayered neural network) with no PCA and all features enabled at 98% accuracy, while the worst performing model was the Ensemble (boosted trees) with PCA enabled and only 6/10 features at 62% accuracy.

Below, we will look at the top performing model (a trilayered neural network) with 98% accuracy with no PCA and all 10/10 features used and its prediction scatterplot, confusion matrix, ROC-AUC, and PR-AUC. We will compare these same graphs to a lower performing model towards the end of the list, an ensemble of RUSBoosted Trees with PCA enabled and 6/10 features selected.

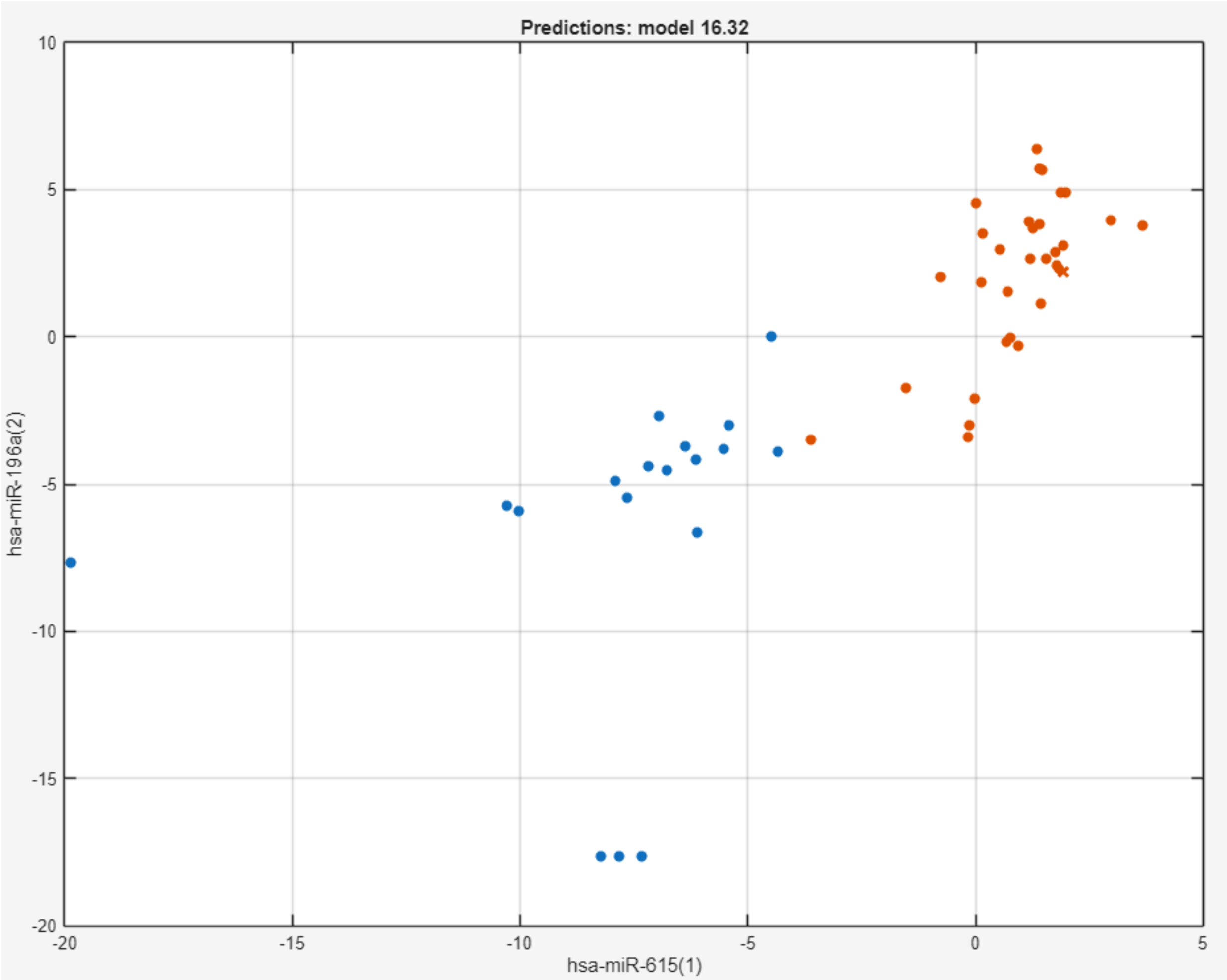

Figure 25. Predictions scatterplot of the best performing trilayered neural network 16.32 of 98% accuracy and its location clustering. Foregut is in blue and midgut is in red.

We can see very good grouping and clustering compared to the model's prediction in this feature space of hsa-miR-615 and hsa-miR-196a with minimal overlap, meaning very good patterns were found from the miRNA features to separate the 2 tumour location classes.

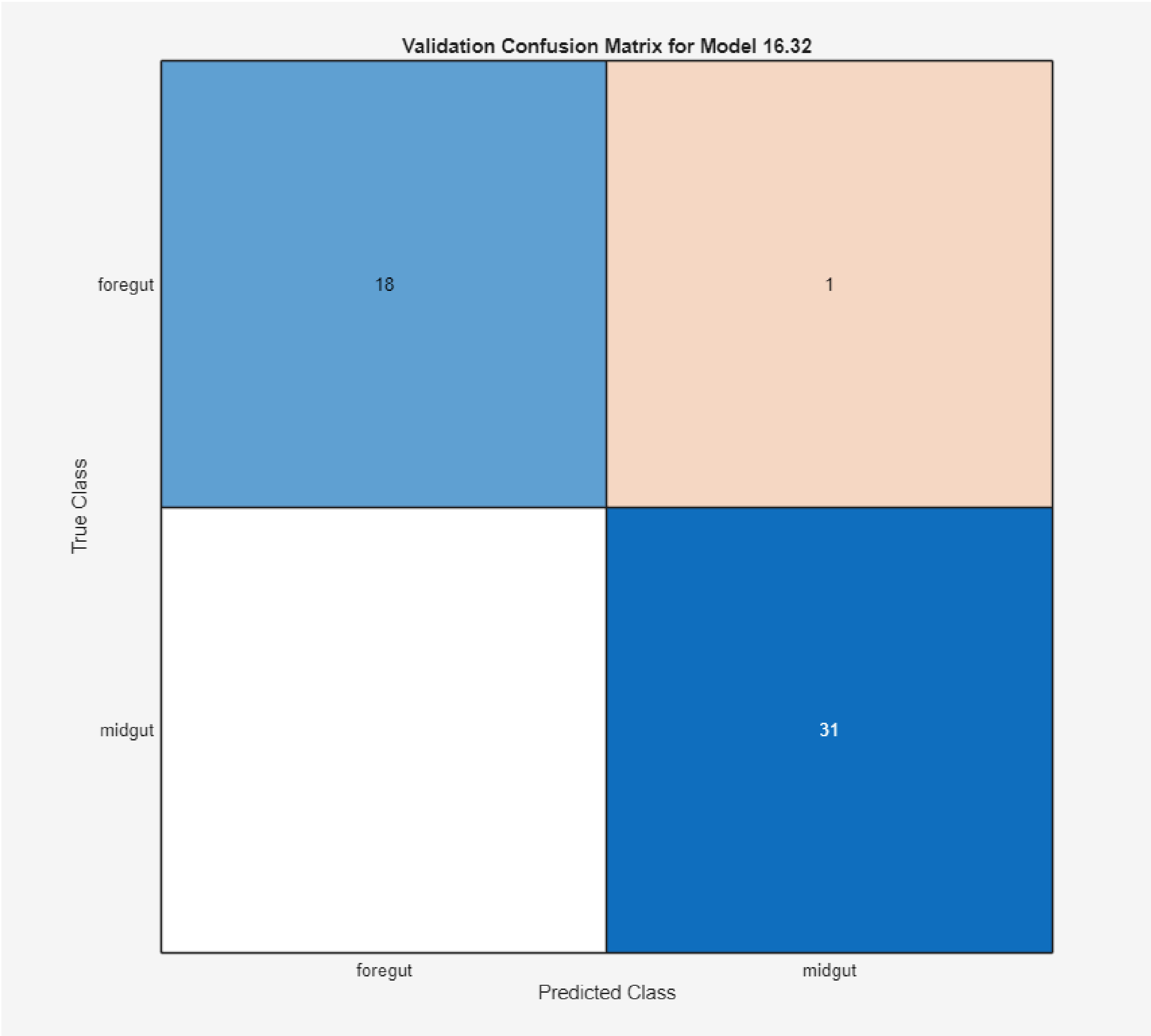

Figure 26. Validation Confusion Matrix of the Top Neural Network Model.

This confusion matrix shows near-perfect classification performance with 49 out of 50 samples correctly classified (18 foregut correctly identified, 31 midgut correctly identified). Only one foregut sample was misclassified as midgut (false positive), while there were zero false negatives (no midgut samples misclassified as foregut). This translates to 98% validation accuracy, with exceptional specificity for midgut detection (100%) and strong sensitivity for foregut detection (94.7%).

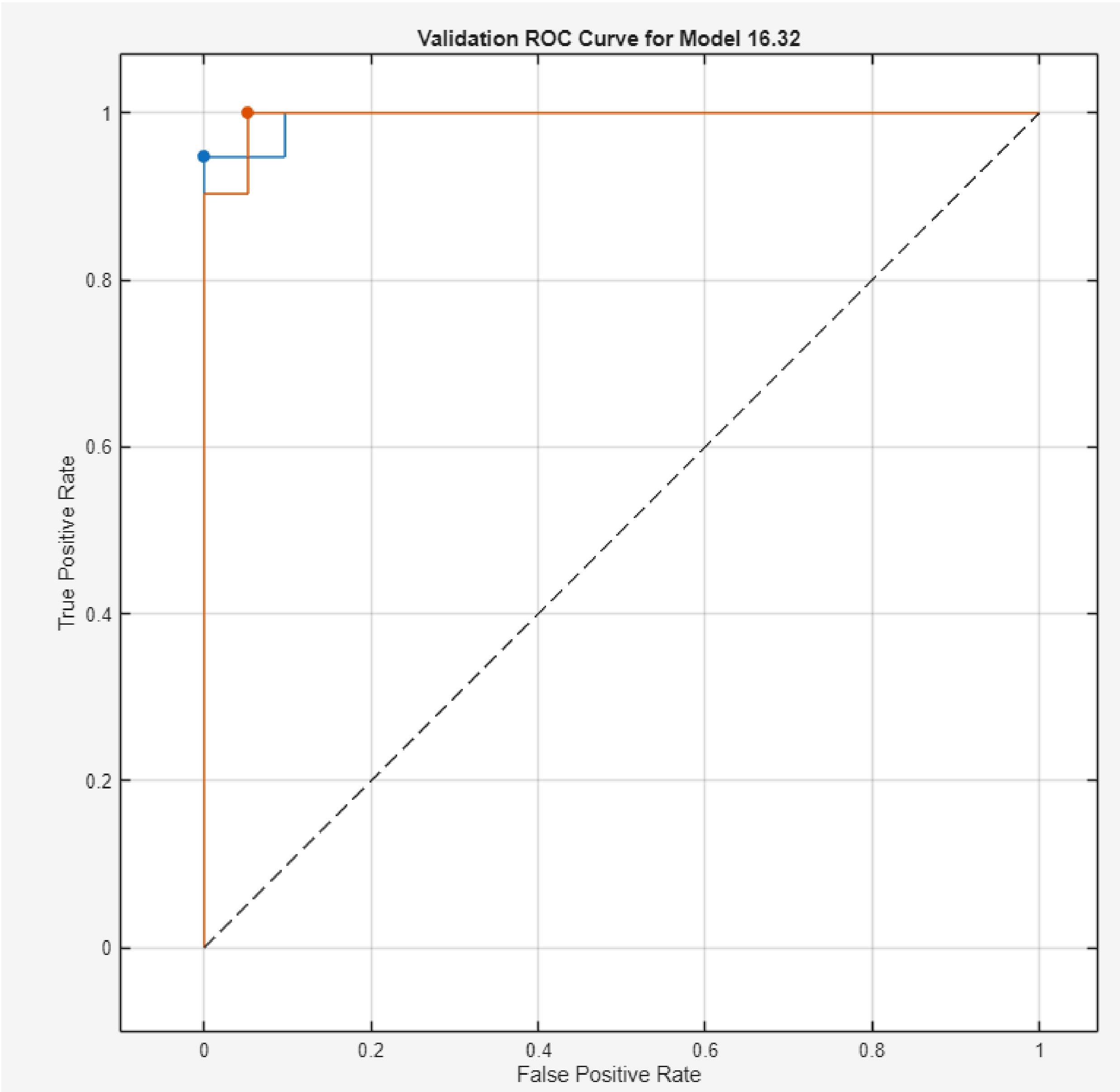

Figure 27. Receiver Operating Characteristic (ROC) Curve for Top Neural Network.

The ROC-AUC is near meaning, meaning the closer it is to 1.0, the stronger the model's ability to distinguish between foregut and midgut tumours across all classification thresholds. The AUC for the foregut in blue is 0.99491 while the AUC for the midgut in red is also 0.99491.

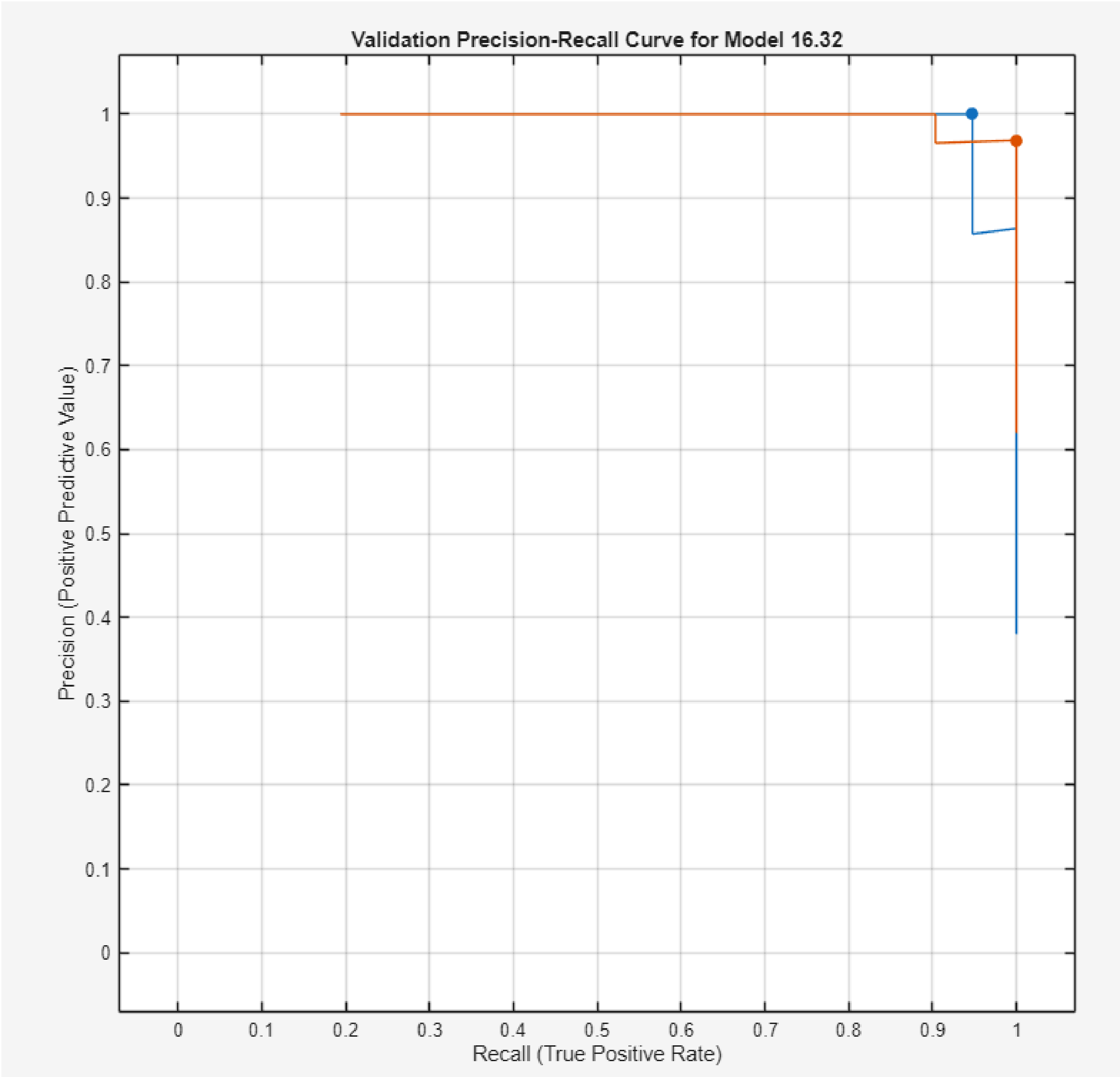

Figure 28. Precision-Recall Curve for Top Neural Network.

The precision-recall curve maintains values near 1.0 across the entire recall range, demonstrating that the model achieves both high precision (proportion of positive predictions that are correct) and high recall (proportion of actual positives correctly identified). The PR-AUC for the foregut in blue is 9.99265 and the PR-AUC for the midgut in red is 0.99682.

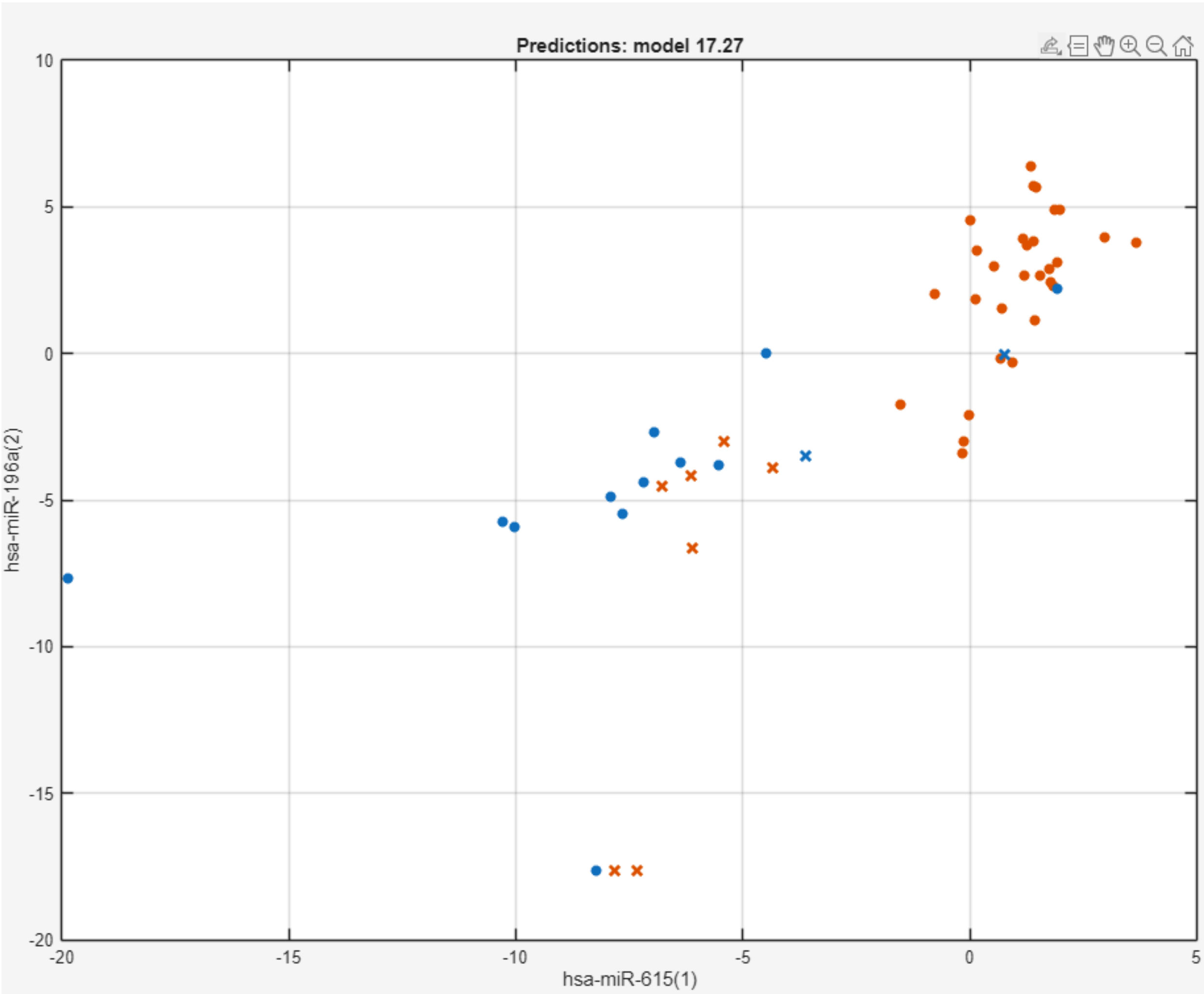

Figure 29. Prediction Scatter Plot for Ensemble RUSBoosted Trees Model with PCA (Model 17.27).

This lower performing Ensemble model at 82% accuracy with PCA enabled and 6/10 features selected has worse separation between foregut in blue and midgut in red. We see overlapping dots and less defined boundary regions, showing that the model struggles more with miRNA expression patterns.

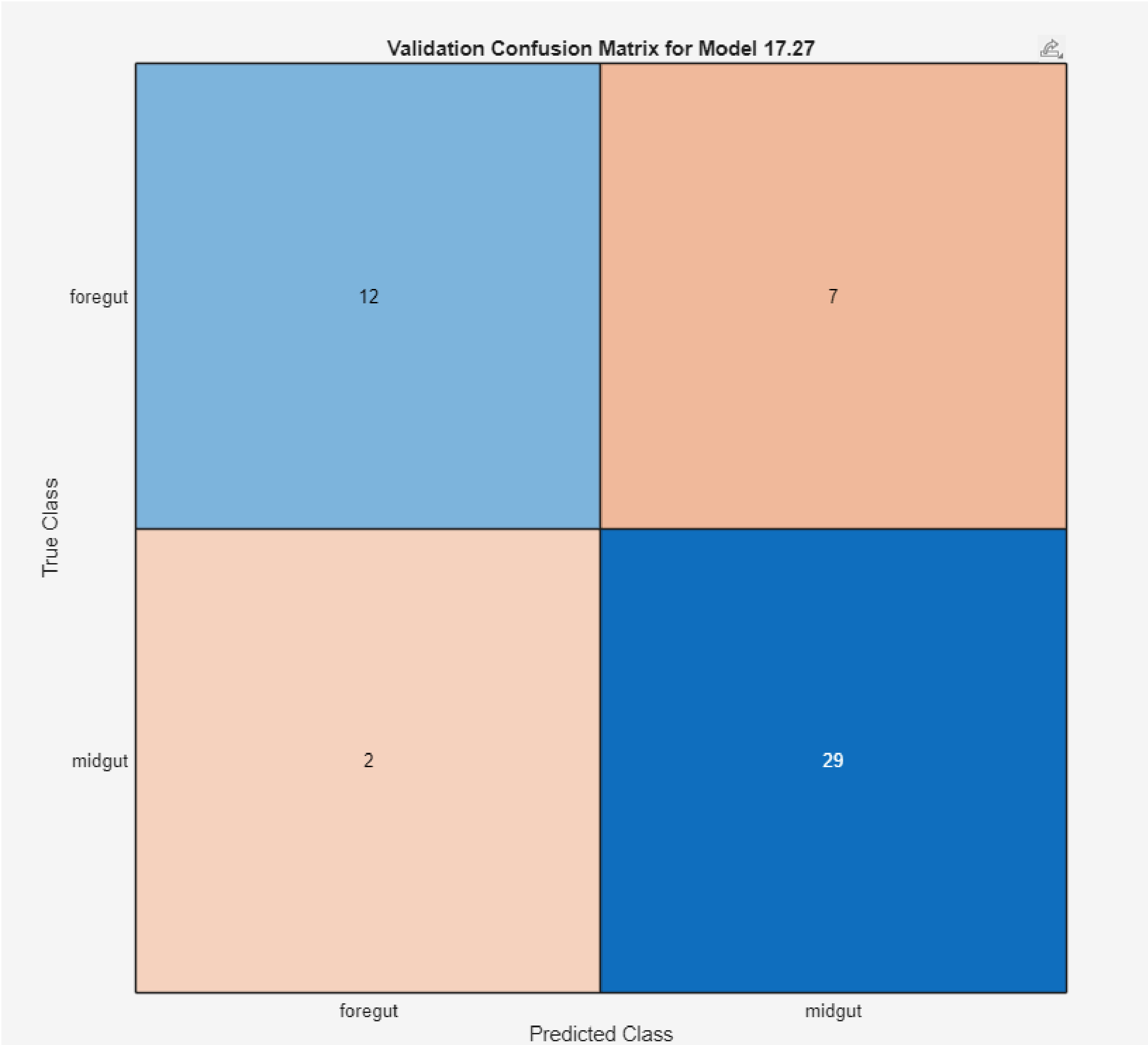

Figure 30. Validation Confusion Matrix for Ensemble RUSBoosted Trees Model with PCA (Model 17.27).

This confusion matrix shows 41 out of 50 samples correctly classified, representing 82% accuracy with considerably more errors than the top model. The model correctly identified 12 foregut samples but misclassified 7 foregut samples as midgut (false positives), while 29 midgut samples were correctly identified with only 2 misclassified as foregut (false negatives). The asymmetric error pattern indicates the model has poor sensitivity for foregut detection (63.2% compared to 94.7% in the neural network) but maintains better specificity for midgut detection (93.5%), likely due to the RUSBoost algorithm's attempt to handle class imbalance through random undersampling.

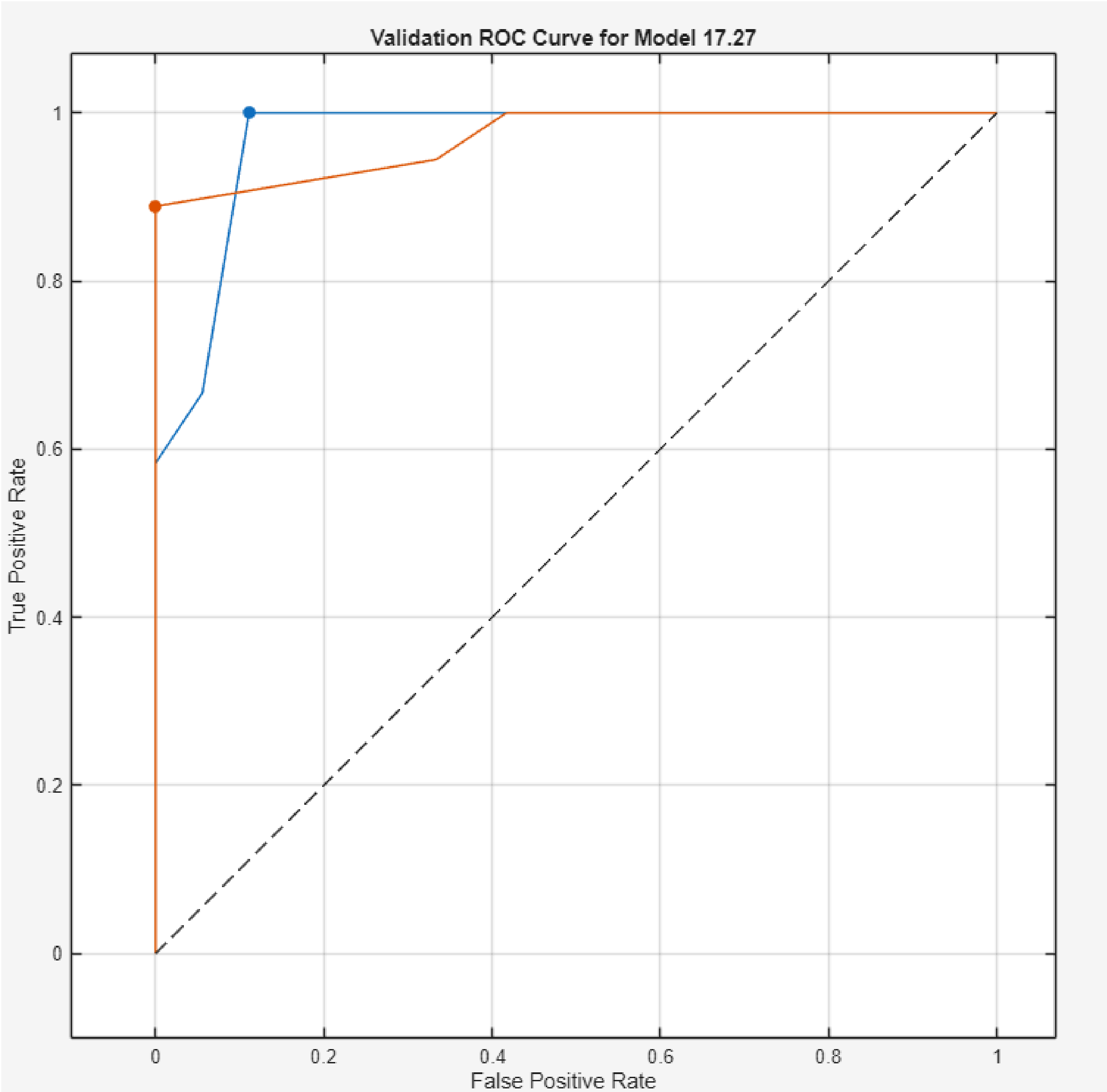

Figure 31. Receiver Operating Characteristic (ROC) Curve for Ensemble RUSBoosted Trees Model with PCA (Model 17.27).

ROC-AUC for the foregut in blue was 0.96991 and ROC-AUC for the midgut in red was 0.96991, demonstrating strong discrimination performance, similar to the top model but still slightly less.

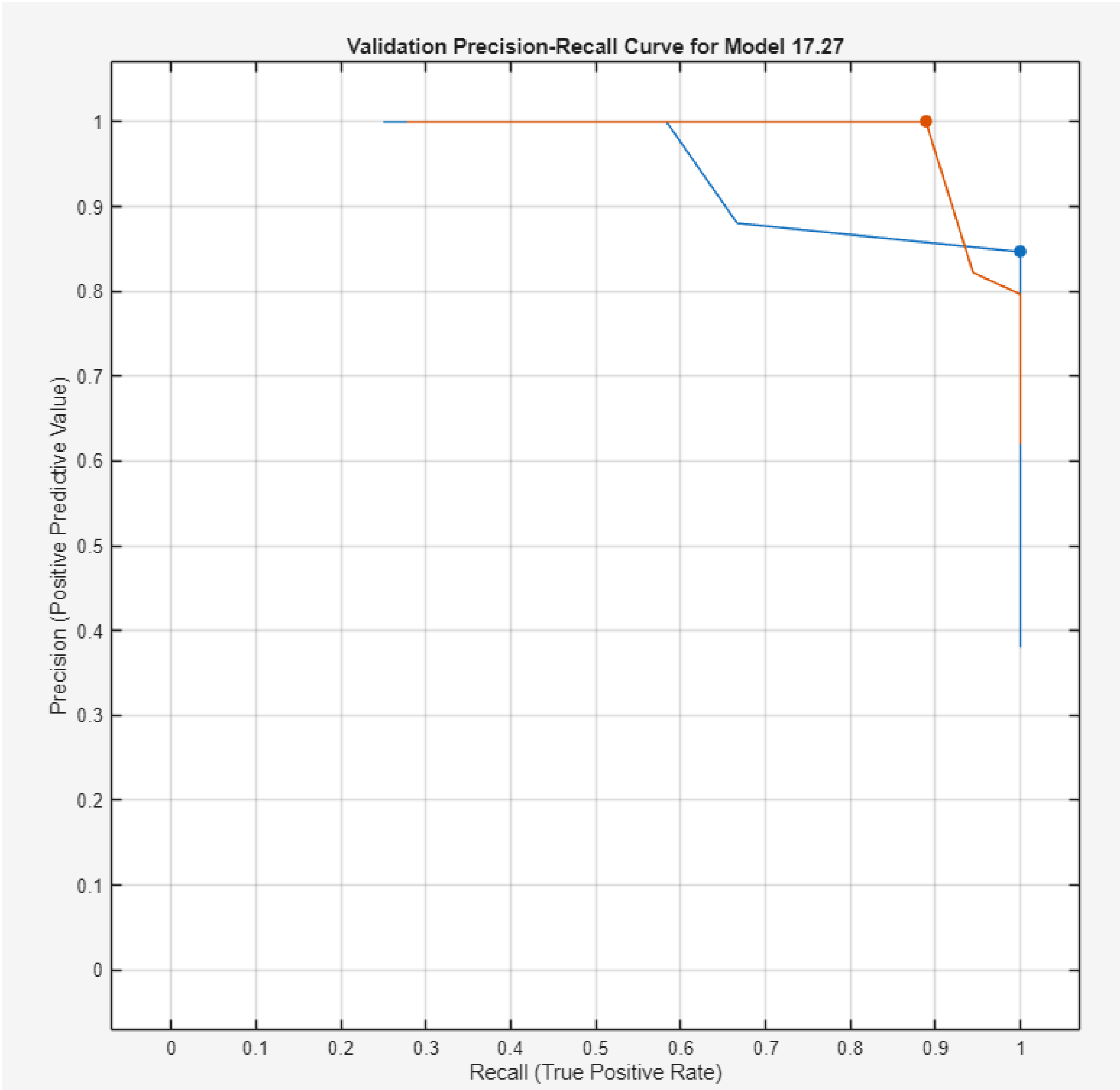

Figure 32. Precision-Recall Curve for Ensemble RUSBoosted Trees Model with PCA (Model 17.27)

The PR-AUC for the foregut was in blue at 0.94949 with the midgut in red at 0.98447, with performance dropping and degrading as the recall increases. Once again, this is not as strong as the neural network.

In summary, the best performing families were the neural networks, discriminant analysis models (linear discriminant achieved 96%), some SVMs with 96% and some KNN models. Moderately performing classifiers (WITHOUT PCA) were Tree-based, achieving 94% accuracy and some Ensemble methods WITHOUT PCA also performed well in the 80s-90s % accuracy. However, with PCA, these ensemble families were the worst, with near 62% accuracy. Other SVM configurations, such as the Fine Gaussian SVM with PCA achieved 72%. Overall, when using PCA or dropping features to 6/10, these models were at the bottom of the accuracy table, suggesting the 10 miRNA features selected by relieff had good non-redundant discriminative information that may have been lost during PCA compression. Models that did not use PCA and all 10 features performed the best across the board.

fscchi2 Feature Selection Method:

In order to identify miRNAs to best discriminate between foregut and midgut tumors, the fscch2 filter-based feature selection method was used. This method evaluates each feature independently of models and ranks them according to statistical relevance score. By measuring the dependency between miRNA expression levels and the categorical location label.

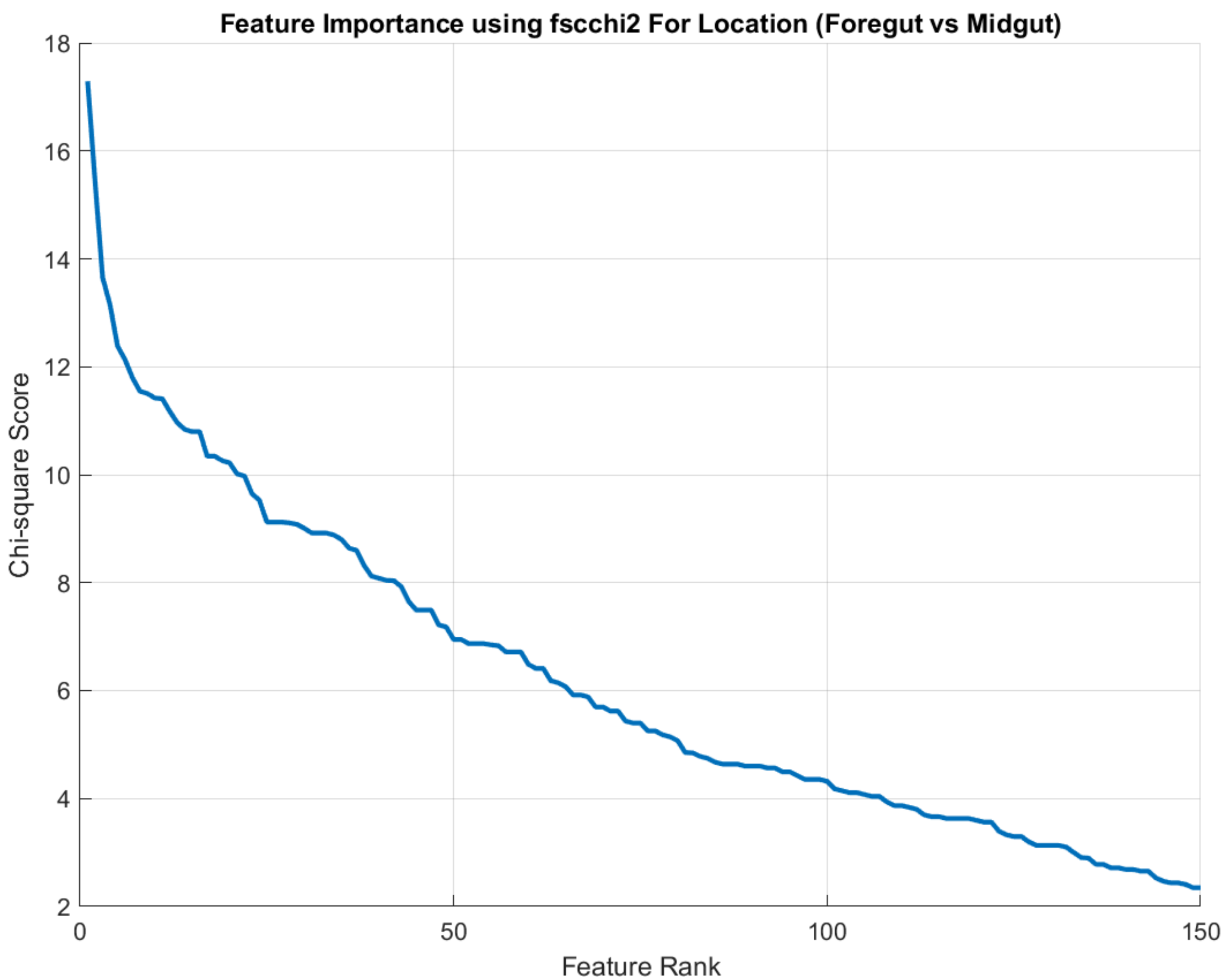

Figure 33. Top 150 Features Ranked Using The fscchi2 Method

In figure 33 only the first 150 features were plotted to more accurately see the elbows and sharp drop off without the tail from the full feature list. From the curve in figure 33 we can see that it exhibits a steep decline after the first 20 features, the chi-square scores start to flatten significantly beyond 20 features, meaning that adding additional features would contribute little more than noise for the machine learning portion. Abstaining from adding too many features also helps with overfitting and a worsening clustering structure.

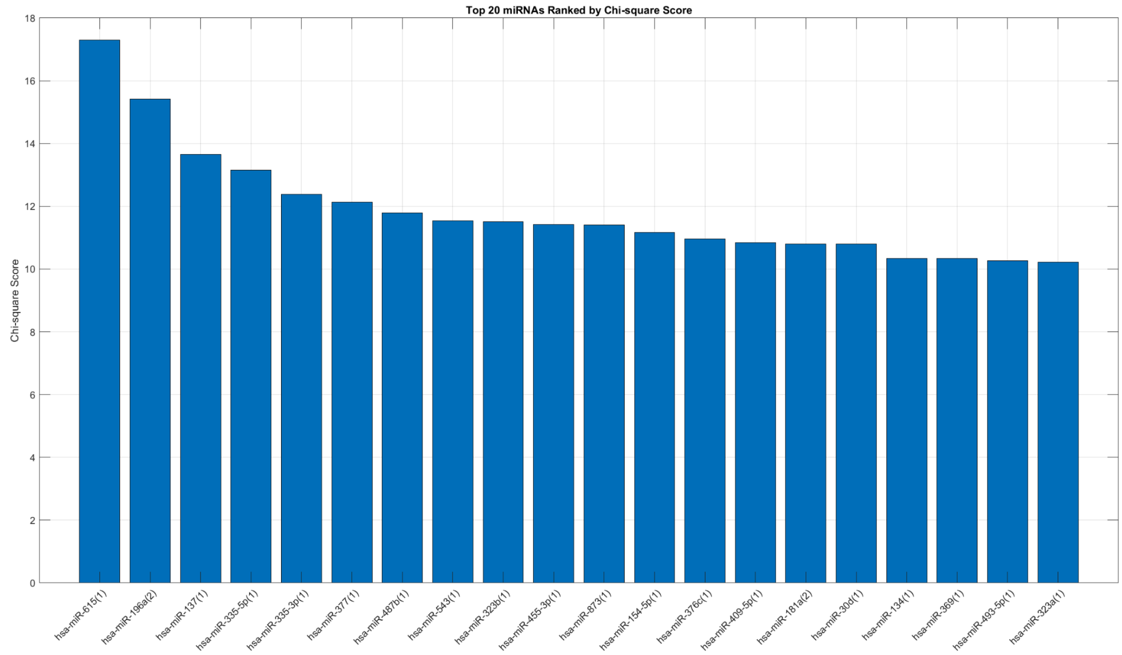

Figure 34. The top 20 Selected Features for Classification Learning

Plot of the top 20 miRNAs, with the strongest statistical relationship with tumor location and were chosen for clustering and classification for machine learning.

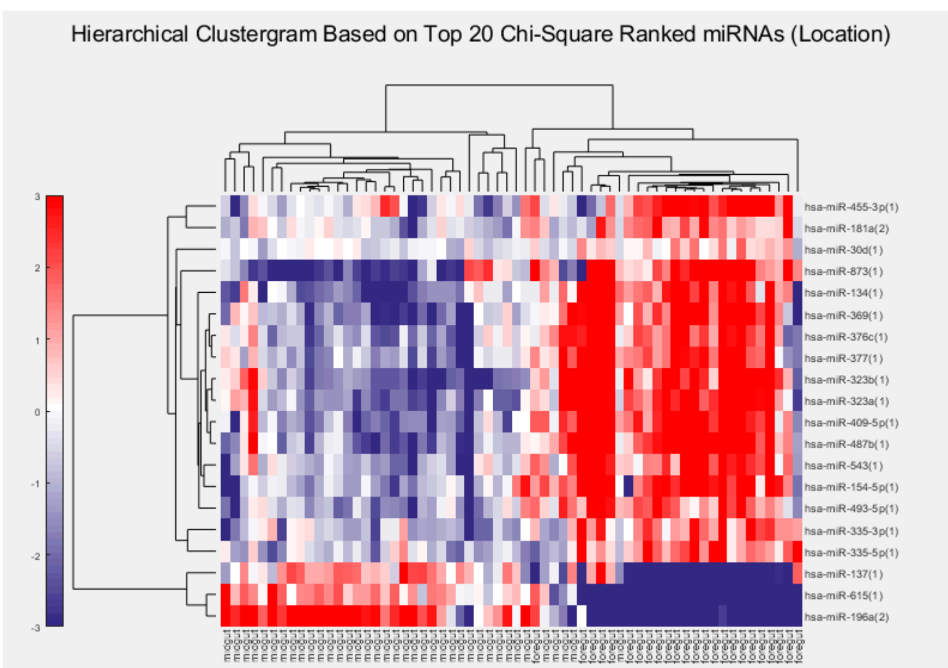

Figure 35. Clustergram of the top 20 features using the fscchi2 method

Compared to the clustering of the full set of features (264), the clustergram of only the top 20 features produced a much clearer separation of samples. The two primary dendrogram branches can be seen, and correspond to foregut and midgut. The blue and red regions are now distinct and indicate clear up or down regulation patterns of the 20 miRNAs. There is significant reduced noise as low information miRNAs have been removed. Using only the top 20 chi-square miRNAs has dramatically improved the clustering quality, confirming that feature selection improves the classification structure for high-dimensional biological data.

The final step was to build supervised classification models using the selected miRNAs. The dataset that was imported into MATLAB's classification learned included the 62 samples and 20 selected miRNAs, and the foregut vs midgut labels. The techniques used were:

- 80/20 train test split
- 5 fold cross-validation for internal validation

Model Type	Accuracy % (Validation)
Efficient Linear SVM	100
Neural Network	100
Neural Network	100
Efficient Logistic Regression	98
Ensemble	98
Ensemble	98
Neural Network	98
Neural Network	98
Neural Network	98
Tree	96
Tree	96
Tree	96
Tree	96
Naive Bayes	96
SVM	96
SVM	96
SVM	96
SVM	96
KNN	96
KNN	96
KNN	96

KNN	96
KNN	96
Ensemble	96
Kernel	96
Kernel	96
Naive Bayes	94
SVM	94
Discriminant	92
Binary GLM Logistic Regression	86

Table 2. fscchi2 Model Performances in Ascending order

Most models were able to achieve a 96-100% validation accuracy with low misclassification and high cross-validated stability across folds.

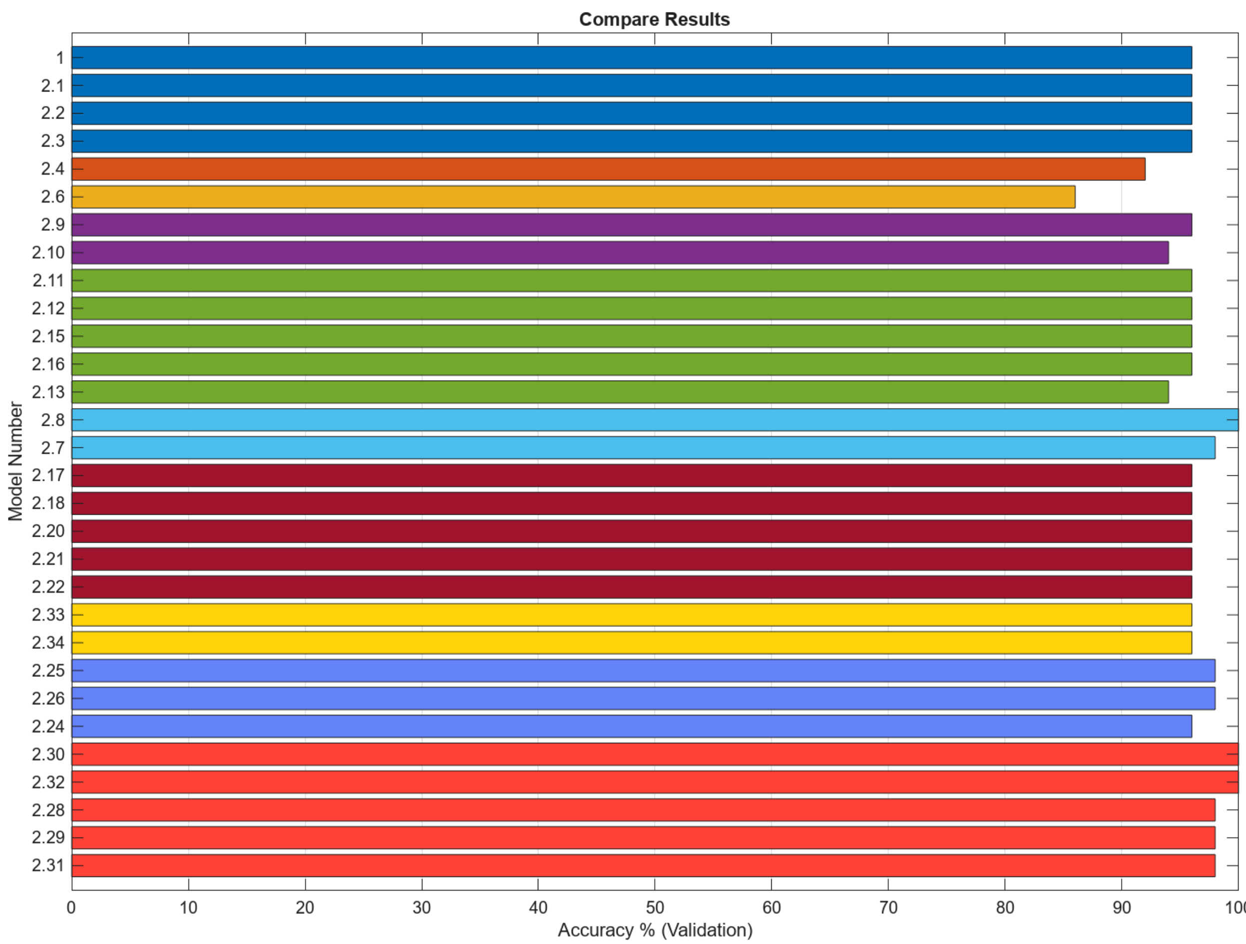

Figure 36. Accuracy Results Grouped by Model Type

Figure 36 shows that several models have achieved 100% accuracy, with the majority of models being in the 98-100% range, with the lowest being 86%. This indicates that the selected feature set has very strong discriminatory information for separating foregut and midgut tumors.

The best performing models were the Neural Networks(light red), Tree (dark blue), SVM(green), while models that performed poorly were Discriminant(orange) and Binary GLM Logistic Regression (light orange). KNNs (dark red) produced slightly lower accuracy, due to their sensitivity to feature scaling and the small sample size.

Using the selected top 20 miRNA features, almost every classifier was able to separate foregut and midgut samples with high accuracy. Confirming that the selection step dramatically boosts both supervised and unsupervised models.

Below is a look at the lowest performing model for the fscchi2 method.

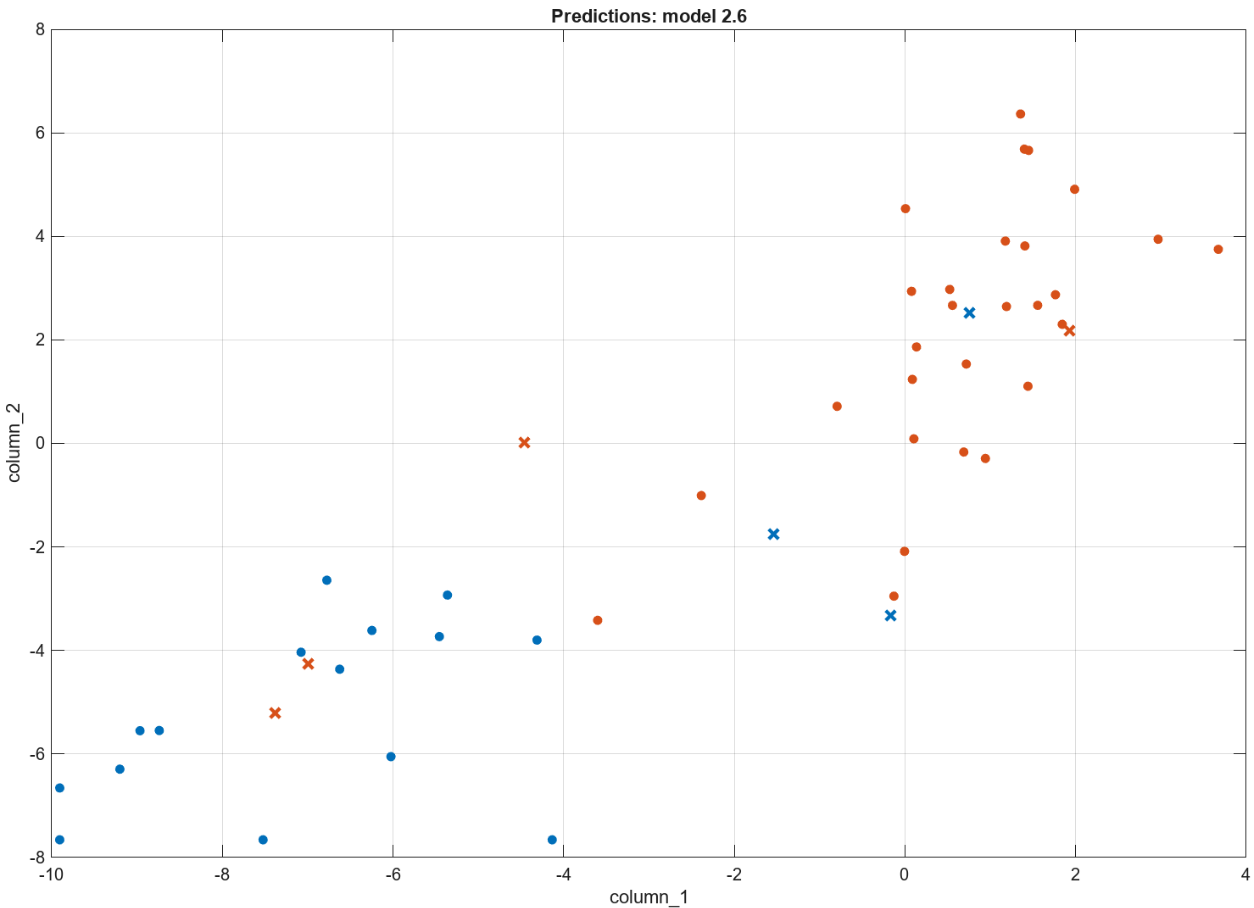

Figure 37. Binary GLM Logistic Regression Scatter Plot

In figure 37 the midgut (orange dots) form a tight grouping on the right side of the plot, but there are several orange X's in the blue region indicating false positives. There are also a few blue X's inside the orange region, indicating false negatives.

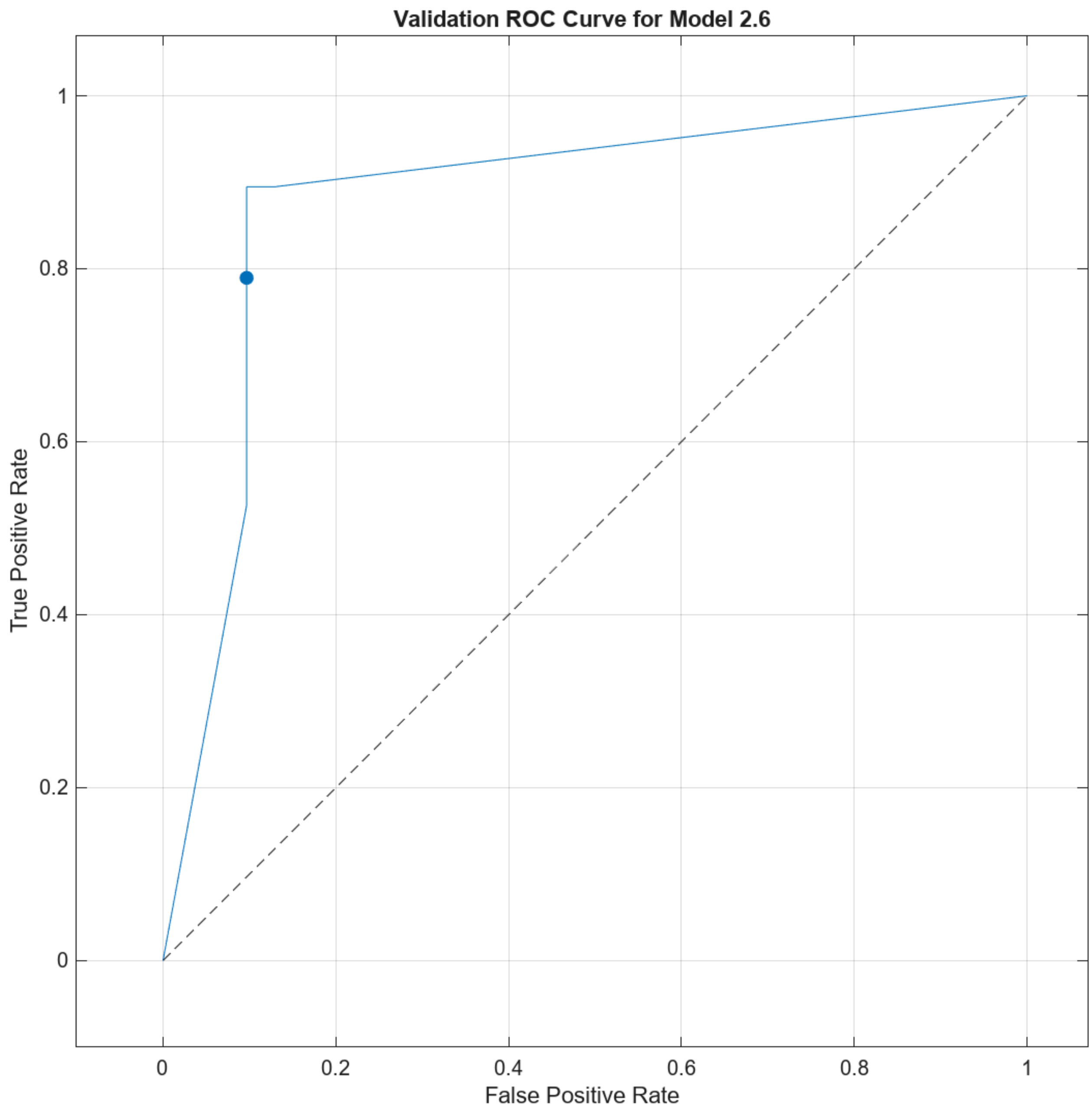

Figure 38. Binary GLM Logistic Regression ROC curve

In figure 38, there is a steep rise at low false positive rates, with gradual flattening as recall increases, and an overall AUC in the 0.90-0.92 range. The logistic regression can distinguish between foregut and midgut better than guessing and has good discriminative capability. The bend in the curve shows the logistic regression's weaknesses when dealing with nonlinear or interaction-driven expression patterns.

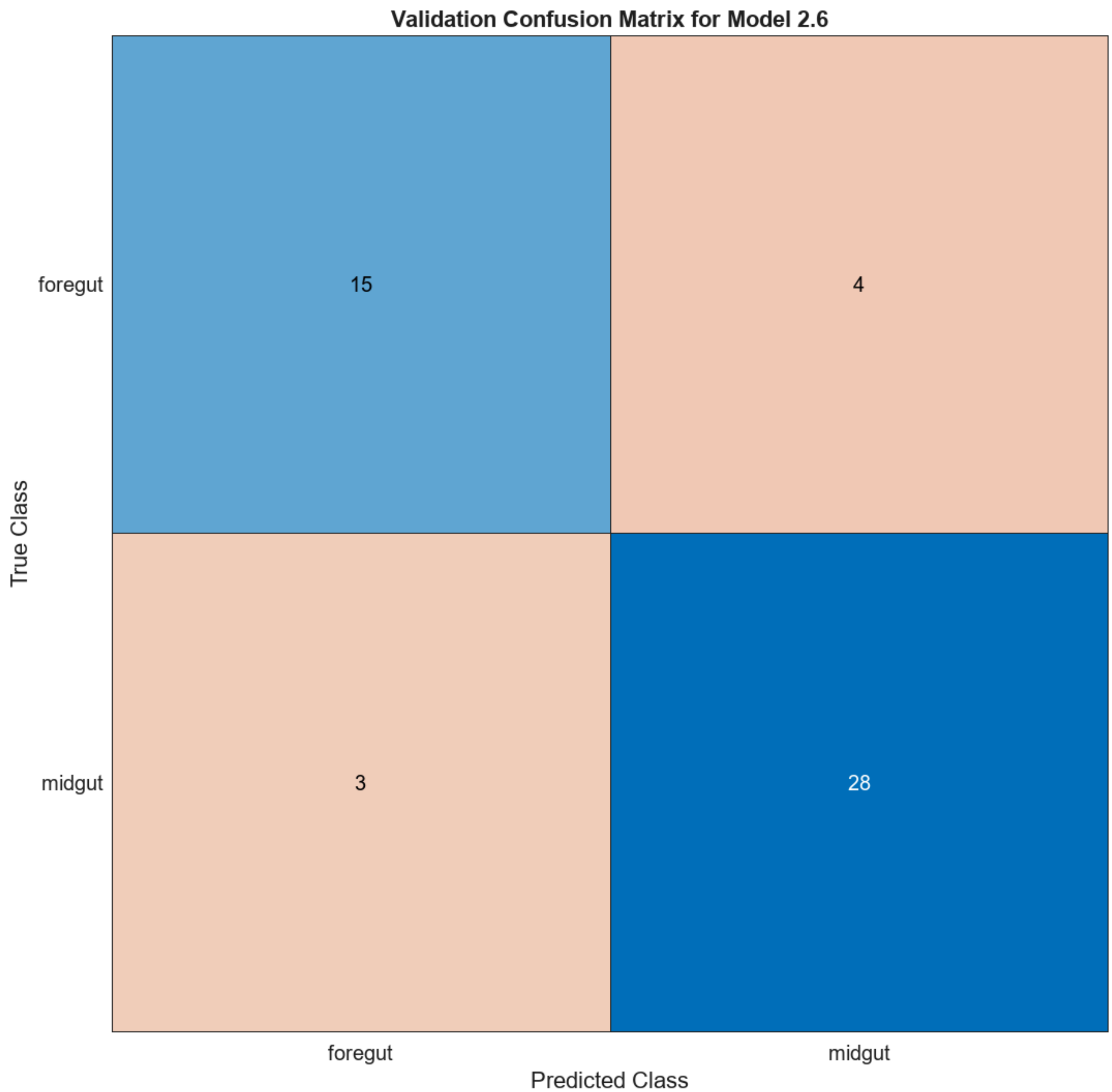

Figure 39. Binary GLM Logistic Regression Confusion Matrix

For figure 39, there is a 86% accuracy (43/50), a foregut recall of 15 / (15+4) = 78.9%, and a midgut recall of 28 / (28+3) = 90.3%. There was 4 forgut that got classified as midgut and 3 midgut that got classified as foregut.

The Binary GLM Logistic Regression model performs better for midgut then foregut samples, this shows balance but imperfect sensitivity, with a tendency to predict midgut when uncertain. The model is more confident in making midgut predictions than foregut.

Feature Selection using fscmrmr

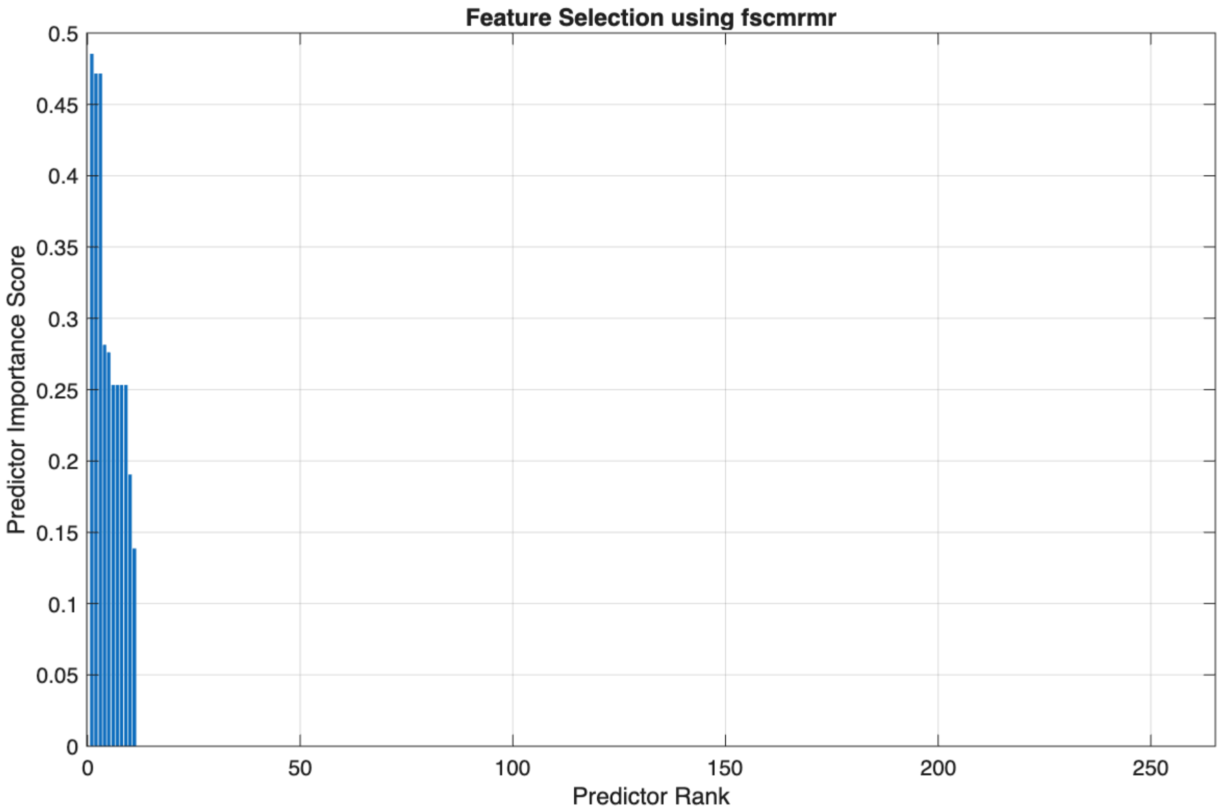

Figure 40. Predictor importance scores of all miRNA features using fscmrmr feature selection

Using fscmrmr, Figure 40 illustrates the predictor importance scores of all miRNA features. Looking at predictor rank, the graph illustrates that out of all 264 miRNA, only a few features have high importance. This can be further illustrated in Figure 41, which removes many of the miRNA with low scores, and looks at the top 50 miRNAs with the highest score. Using this figure, it becomes clear that 11 miRNAs have high importance and there is a large drop after. As such, it would be beneficial to use these 11 miRNA to build a classifier with high discriminatory power. Figure 42 allows for a greater visualization of the predictor importance scores of each of the top 11 miRNA.

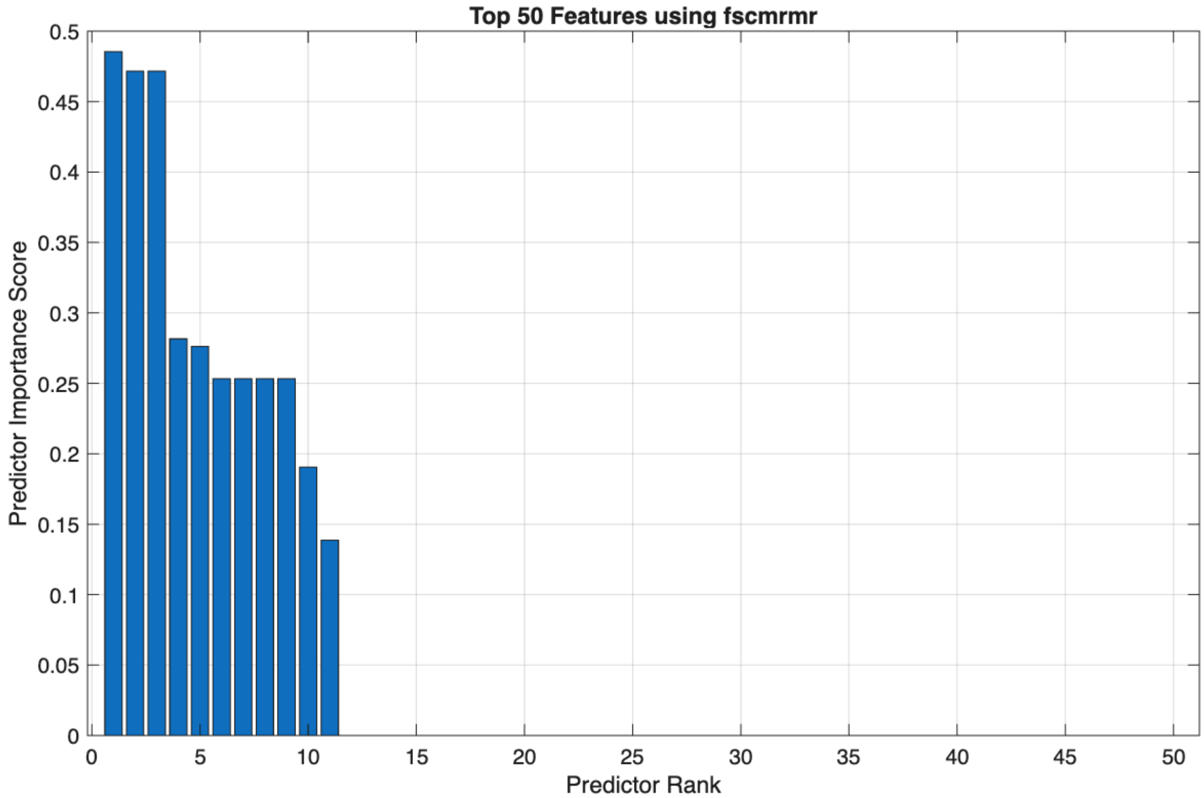

Figure 41. Predictor importance scores of the top 50 miRNA features using fscmrmr feature selection

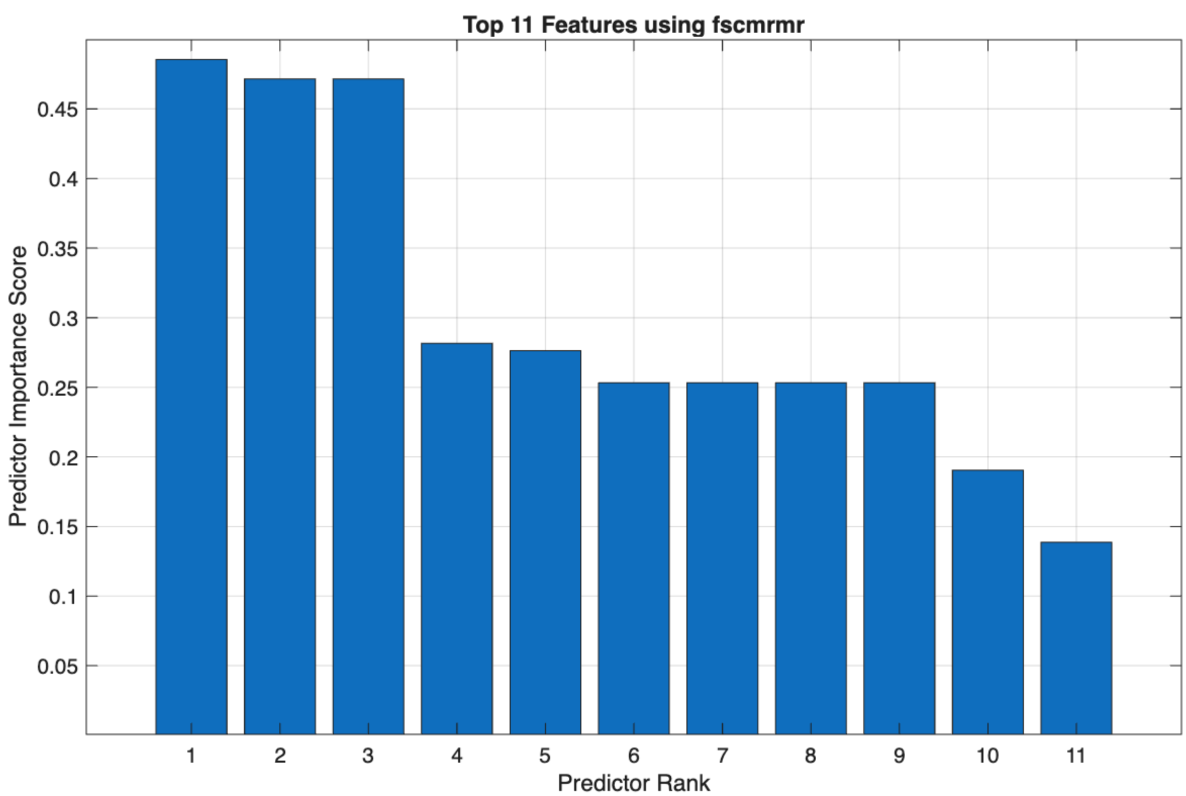

Figure 42. Predictor importance scores of the top 11 miRNA features using fscmrmr feature selection

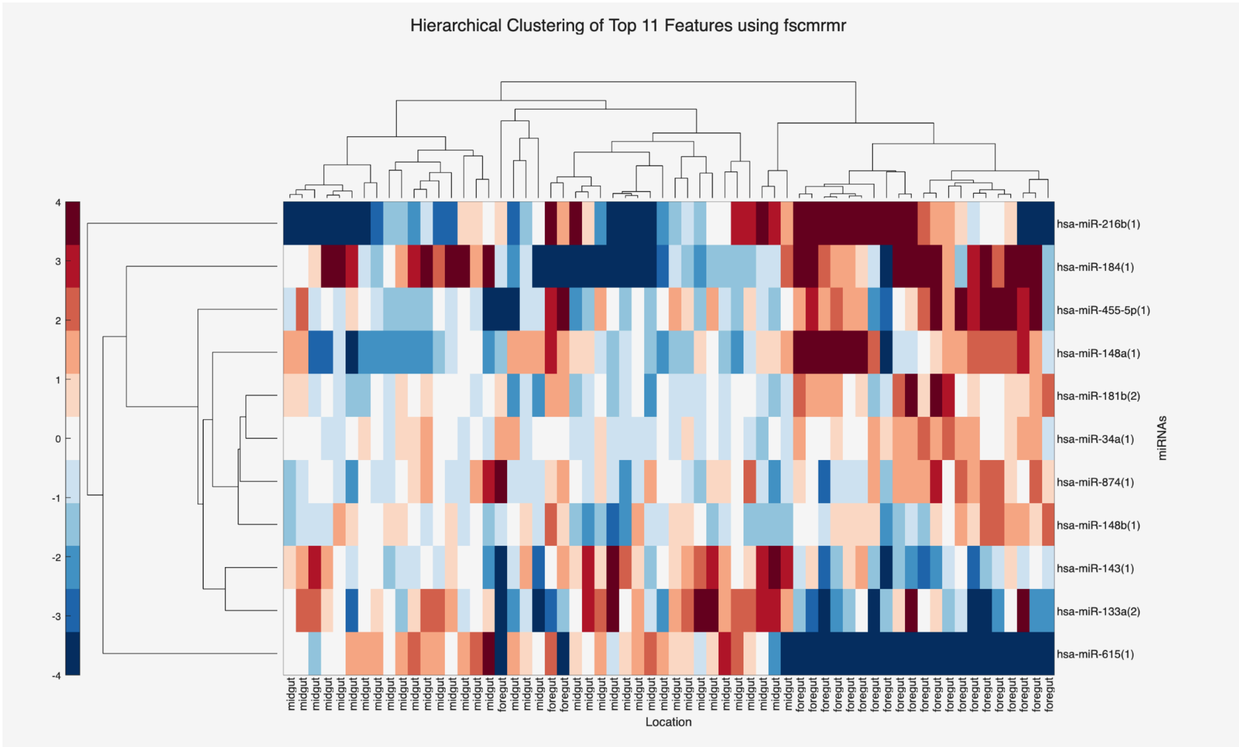

Figure 43. Hierarchical clustering and heatmap of top 11 features selected using fscmrmr

As seen in figure 43 with the hierarchical clustering and heatmap of the top 11 features using fscmrmr, there is some clustering illustrated. miR-615 demonstrates low expression in the foregut. Moreover, miR-216 demonstrates some clustering of high expression in the foregut and low expression in the midgut. Other than these groupings, there is still noise in the dataset. While the method helped to remove many miRNAs that have variable expression from part 2, some of the selected features still have no distinct pattern of high expression or low expression in the foregut or midgut areas. Nonetheless, as no other scores were of significant importance when using fscmrmr, the selected miRNAs will be used to create a classifier through machine learning.

Machine learning was conducted using the top 11 selected features. 5-fold cross-validation, and an 80/20 split of training data and testing data was used for reasons described above in the relieff feature selection method. Using all machine learning models, the validation accuracy summary table is provided below, illustrating the top 20 with the highest accuracy.

Validation Accuracy Table	
Model	Validation Accuracy (%)
Efficient Linear SVM	100
SVM (Medium Gaussian)	100
Kernel (SVM)	100
Kernel (Logistic Regression)	100
Linear Discriminant	98
Quadratic Discriminant	98
Efficient Logistic Regression	98
SVM (Quadratic)	98
SVM (Cubic)	98
KNN (Cosine)	98
Ensemble	98
Neural Network (Medium)	98
Neural Network (Wide)	98
Tree (Fine)	96
Tree (Medium)	96
Tree (Coarse)	96
Binary GLM Logistic Regression	96
Naive Bayes	96
SVM (Linear)	96
SVM (Coarse Gaussian)	96
Ensemble (Bagged Trees)	96
Ensemble (Subspace KNN)	96
Neural Network (Narrow)	96
Neural Network (Bilayered)	96

Naive Bayes (Kernel)	94
KNN(Fine)	94
KNN (Medium)	94
Neural Network (Trilayered)	94
KNN (Cubic)	92
KNN (Weighted)	92
SVM (Fine Gaussian)	60
KNN (Coarse)	60
Ensemble (Boosted)	60
Ensemble (RUSBoosted Trees)	60

Table 3. Validation accuracy of the classification models using fscmrmr method

Table 3 illustrates that different classification models validation accuracy using the fscmrmr method. 4 methods have the highest validation accuracy, with a value of 100%. Three of these classification models fall into the support vector machine (SVM) family. Below, we will explore a high performing (100% validation accuracy) model, Efficient Linear SVM, with no PCA and all 11 features used, focusing on the confusion matrix, ROC, and scatterplot.

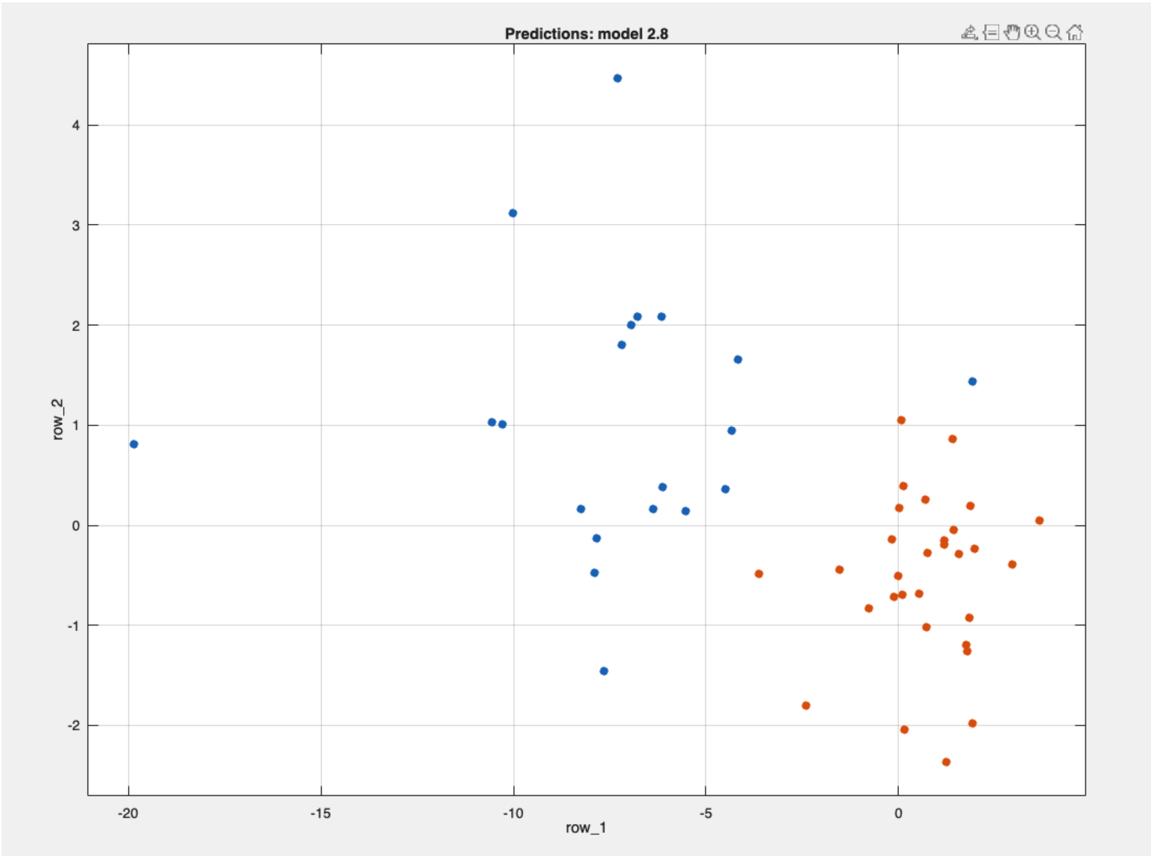

Figure 44. Scatterplot for Efficient Linear SVM (without PCA). Midgut samples are red and foregut samples are blue.

Using Efficient Linear SVM (without PCA) demonstrates good clustering of midgut cancer samples. Nonetheless, the foregut cancer samples demonstrate some outliers located away from the main cluster.

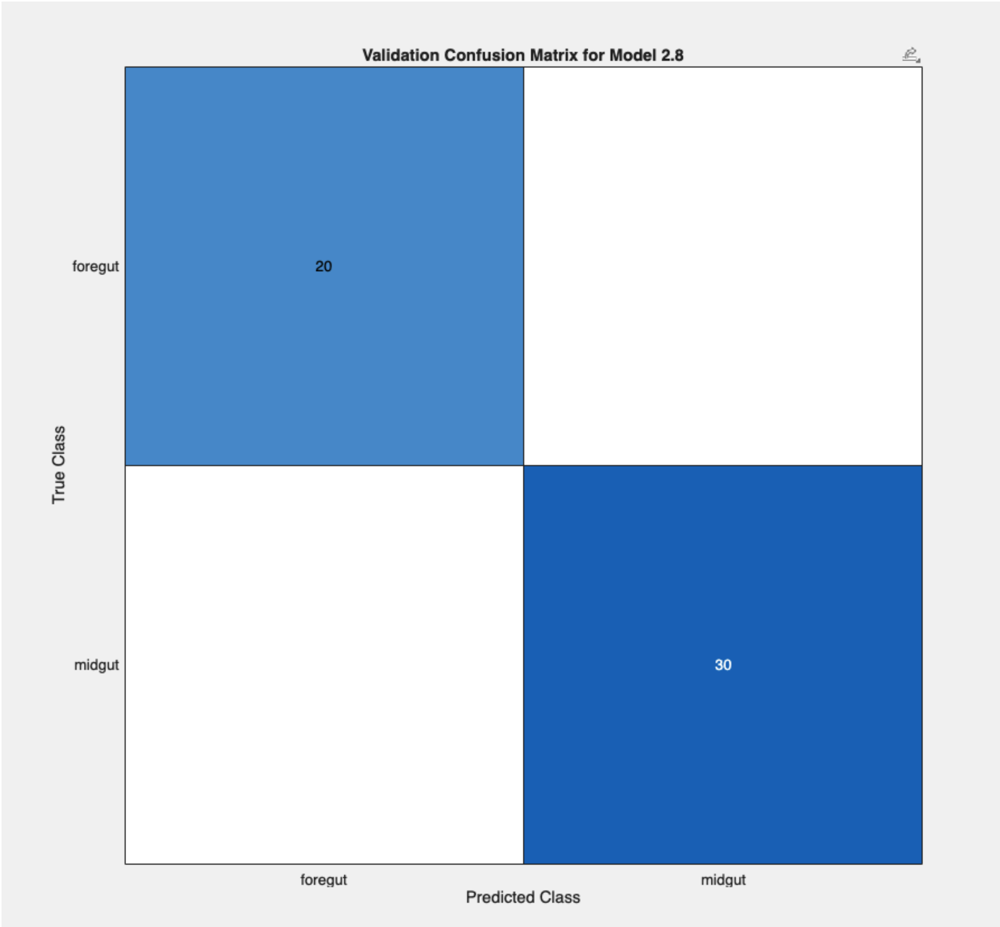

Figure 45. Validation Confusion Matrix for Efficient Linear SVM.

This confusion matrix illustrates perfect classification performance, with all 50 samples classified as midgut or foregut. This helps to illustrate the 100% validation accuracy that was identified when running the different classification models. As such, this illustrates a high specificity and sensitivity of 100% for both foregut and midgut detection. Figure 46 illustrates a ROC curve, further supporting the high validation accuracy with a false positive rate of 0, and a true positive rate of 1. Moreover, the AUC for both foregut and midgut is 1 respectively.

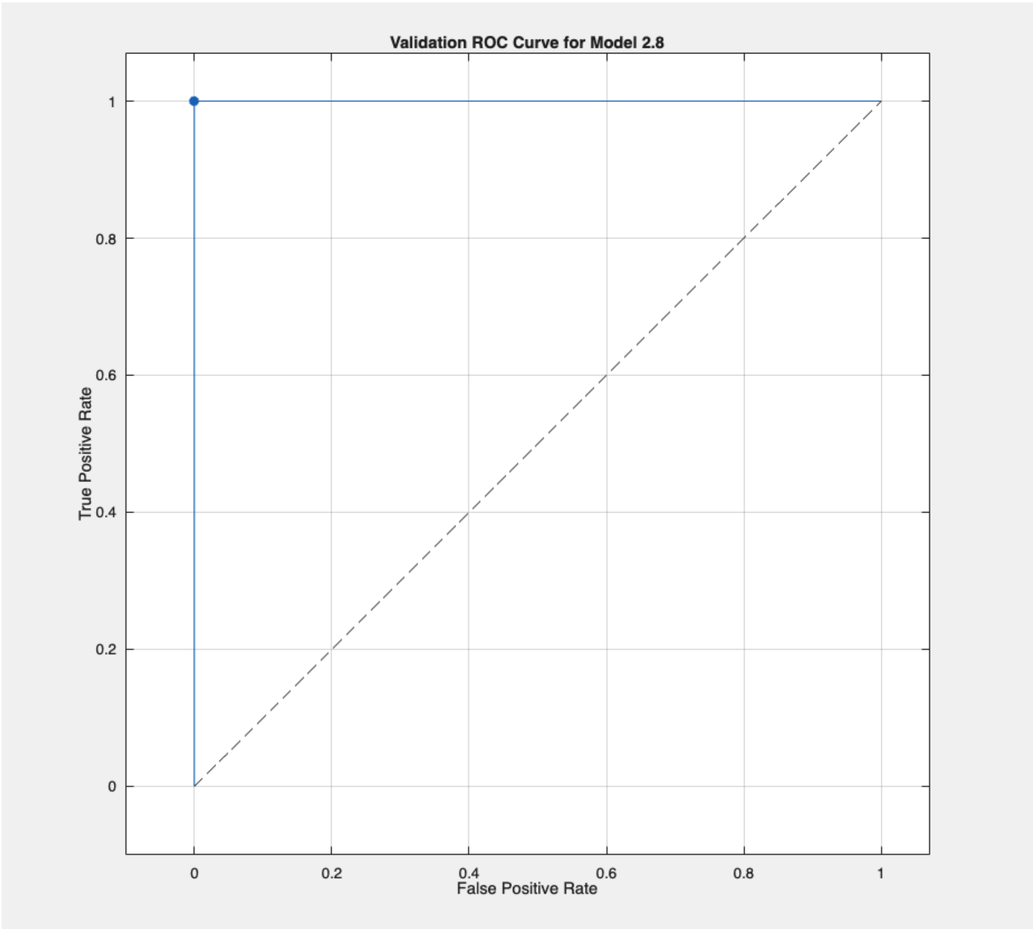

Figure 46. Validation ROC curve for Efficient Linear SVM

The best performing classification models are SVM, with three models showing 100% validation accuracy. Other good performing models included discriminant models, three of which have a 98% validation accuracy, and decision tree classifier models, with three models having a 96% validation accuracy.

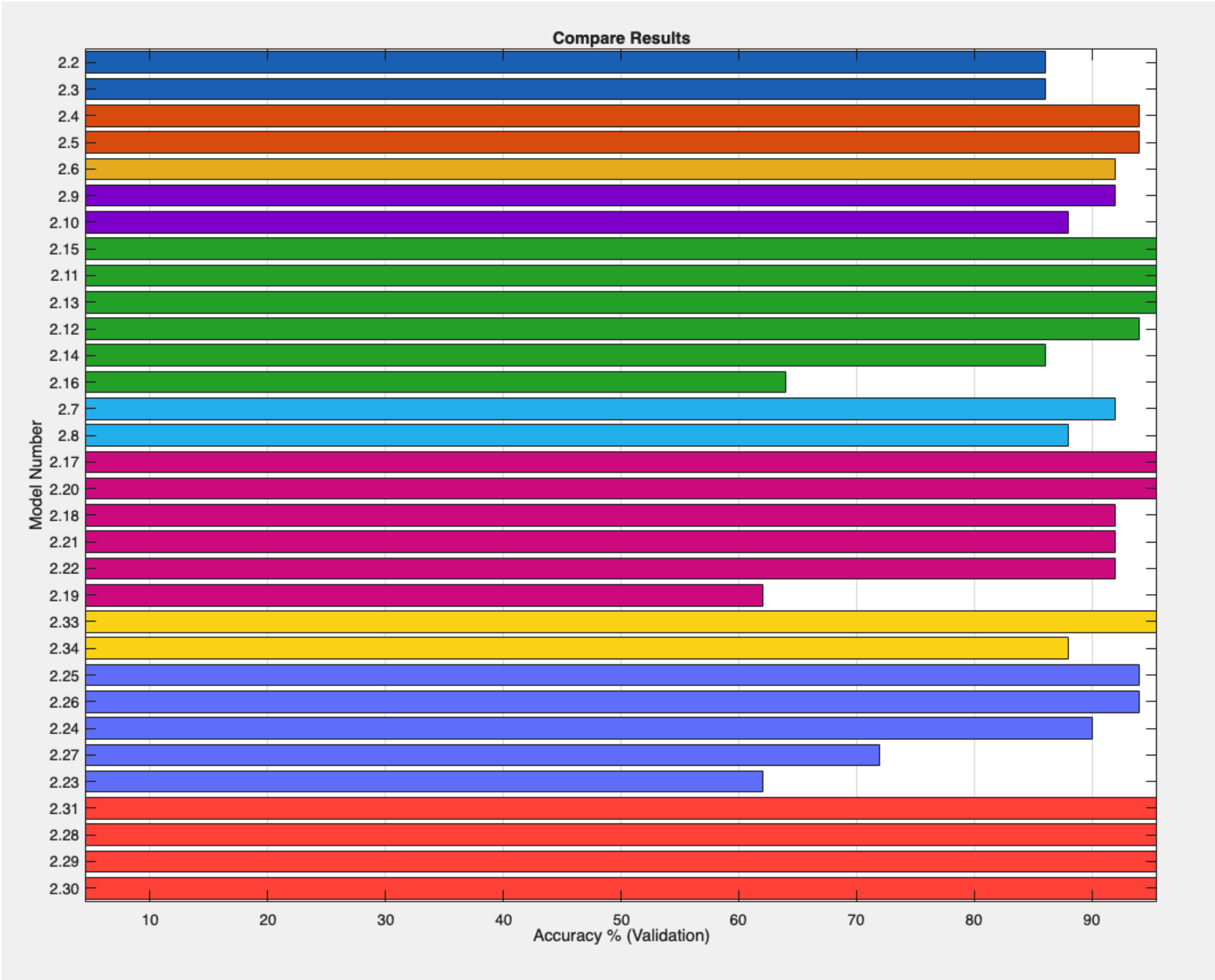

Figure 47. Comparison of the validation accuracy across different classification methods when enabling PCA.

As seen in figure 47, the validation accuracy of the classification models decreases with enabling PCA, with numerous validation accuracies for different models falling in the range between 60% - 90% in comparison to when PCA was not enabled (Table 3). Using PCA resulted in some of the discriminatory abilities of different models being lost.